Windows Server: Administration Basics

January 2025



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Agenda (1)

Introduction to Windows Server
- Services
- Applications
- Security Concepts

Methods and tools for administration
- Management tools
- Local and remote administration

Graphical administration tools
- Microsoft Management Console
- Server Manager
- Windows Admin Center

Command line administration tools
- Command line hosts
- Windows Terminal
- CMD.EXE
- PowerShell

Common administrative tasks
- Managing role and features
- Managing services
- Reviewing events
- Configuring firewall

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Agenda (2)

Manual setup and first configuration
- Setup
- Desktop experience
- Core experience
- Sconfig

Managing users, computers, and groups
- Local users
- Active Directory basics
- Domain users
- Groups
- Computers

Managing local storage
- Disks
- Partitions
- Volumes
- NTFS
- ReFS
- RAID
- Mount points
- Links

Introduction to PKI
- Applications
- Algorithms
- Certificate management

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Agenda (3)

Managing permissions and shares
- Permissions
- Shares
- Offline files
- Volume Shadow Copies

Introduction to Group Policies

Performance Monitoring
- Counters
- Live monitoring
- Data collection

Licensing
- Windows Server editions
- Licensing
- Virtualization rights
- Azure hybrid use benefit

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Introduction to Windows Server

Services, applications, security concepts



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What is a server?

In computing, a server is a piece of computer hardware or software (computer program) that provides functionality for other programs or devices, called “clients”. This architecture is called the client–server model.

Servers can provide various functionalities, often called "services", such as sharing data or resources among multiple clients, or performing computation for a client.

A single server can serve multiple clients, and a single client can use multiple servers.

Server (computing) - Wikipedia

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How does a server work?

Server applications are implemented as services

In other operating systems, services are called daemons

Services run in the background without a user logged in

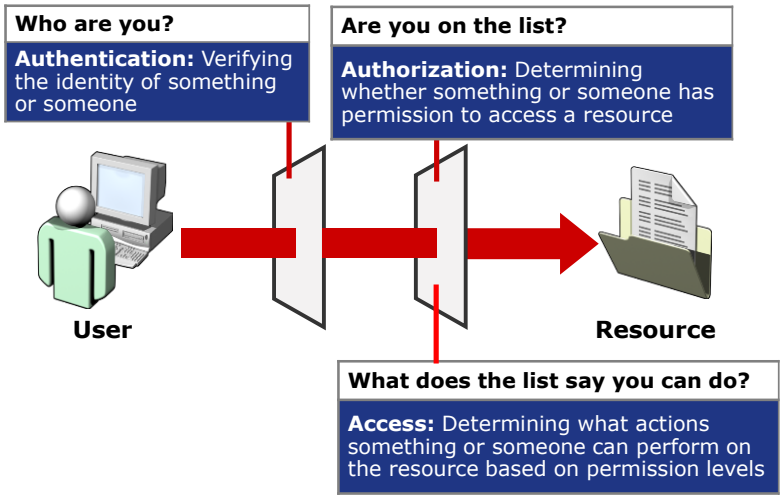
Windows service – Wikipedia

```
While (true) {
    request = checkForRequest()
    if (request != null) {
        doSomething(request)
    }
}
```

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Identity, authentication and authorization



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Identity

Set of data that uniquely describes a person or an object, sometimes referred to as subject or entity, and contains information about the subject's relationships to other entities

Identities are saved in an identity store known as a directory database

In AD DS, identities are called security principals

In an AD DS domain, identities are represented uniquely by the SID

Identities are used mainly to access the resource

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Authentication

The process that verifies a user's identity through credentials

For credentials, at least two components are required

Two types of authentication:

- Local (interactive) log on to a local computer
- Remote (network) log on for access to resources on another computer

Stand-alone authentication

- Users are authenticated by local SAM

Joining a computer to a domain

- Users are authenticated by a domain

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Authorization

The process that determines whether to grant or deny a user a requested level of access to a resource

Three components are required for authorization:

- Resource
- Access request
- Security token

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Identity infrastructure

An identity Infrastructure should:

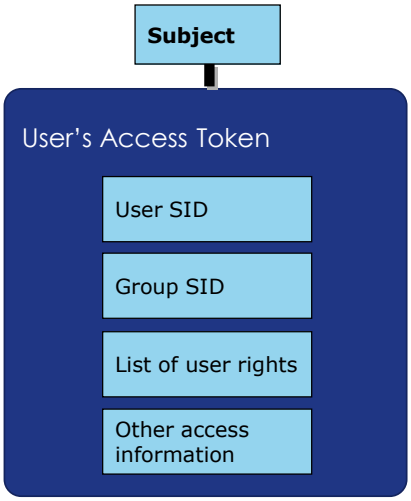
- Store information about users, groups, computers, and other identities
- Authenticate an identity

  Kerberos authentication used in an Active Directory domain provides SSO, and users authenticate only once
- Control access
- Provide an audit trail

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What Are Access Tokens?

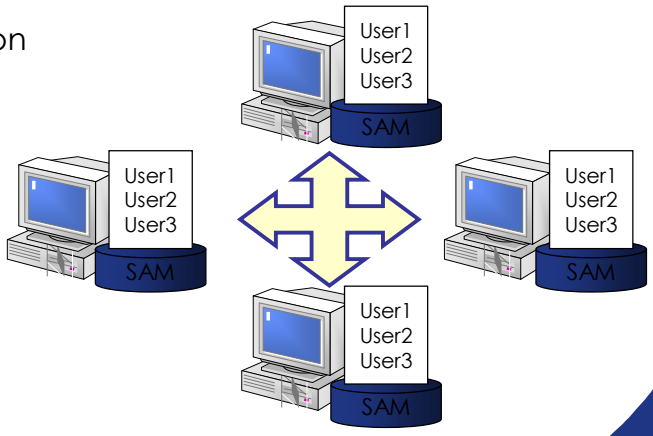


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Workgroup

- Peer-to-peer network
- No centralized administration



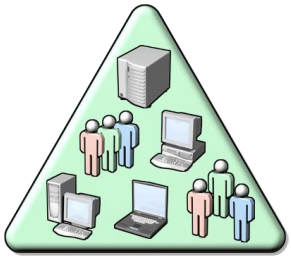
14

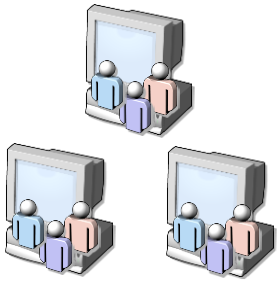
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Directory Service

A directory service is both the directory information source and the service that makes the information available and usable.

Centralized Administration	Dispersed Administration





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Applications of Windows Server and associated courses

Authentication (Domain Controller)	• WINADI: Windows Server Active Directory and identities • WINGRL: Windows Server group policies
Network services (DHCP, DNS, …)	• WINNW: Windows Server network services • WINRA: Windows Server remote access
Hypervisor (virtualization platform)	• WINVH: Windows Server virtualization and high availability
File server	• WINGLA: Windows Server administration basics • WINFSM: Windows Server file server management
Desktop Virtualization (Remote Desktop Services, Virtual Desktop Infrastructure)	• WINRA: Windows Server remote access
Application server (Web Apps, databases, e-mail, …)	• SIIS: Microsoft IIS Web Platform Administration Essentials • IIS: Internet Information Services Management

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More information

Course [WINFAD: Adanced administration and deployment]
[What's new in Windows Server 2022 | Microsoft Docs]
[Features removed or no longer developed starting with Windows Server 2022 | Microsoft Docs]

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Methods and tools for administration

Management tools, local and remote administration



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Administrative tools

Graphical tools	Web apps	Command line tools
• Microsoft Management Console (MMC) • Server Manager • Other tools	• Windows Admin Center	• CMD.EXE • PowerShell

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Local and remote administration

Local Administration
- On the local server console
- Using a Remote Desktop connection

Remote administration
- Windows Admin Center
- PowerShell remoting
- Remote Server Administration Tools

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Graphical administration tools

Microsoft Management Console, Server Manager, Windows Admin Center

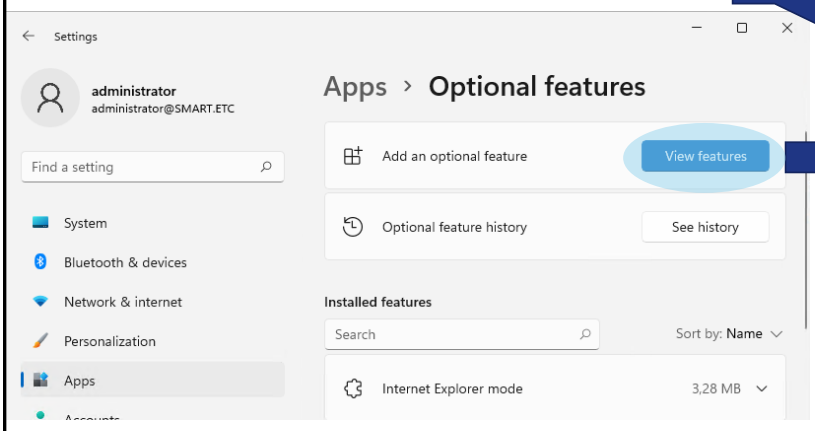


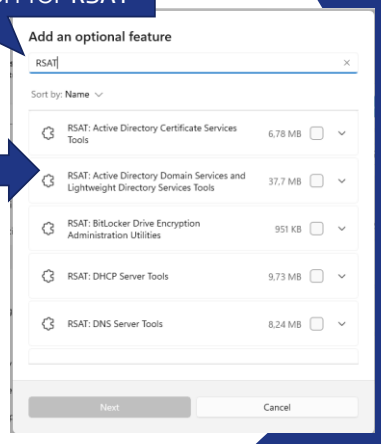
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Installing Remote Server Administration Tools

ETC

Search for RSAT





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Practice: Install Remote Server Administration Tools

Install the Remote Server Administration Tools for Active Directory Domain Services, File Services, and the Server Manager on CL1.

Required time: 5 minutes



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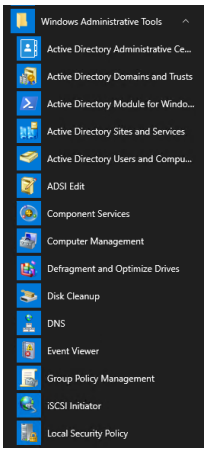
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Microsoft Management Console (MMC)

Framework for snap-ins, which provide the actual functionality

Most administrative tools in the **Windows Administrative Tools** group of the start menu are pre-configured MMC consoles

Additional pre-configured MMC consoles are available, e.g., `certlm.msc`



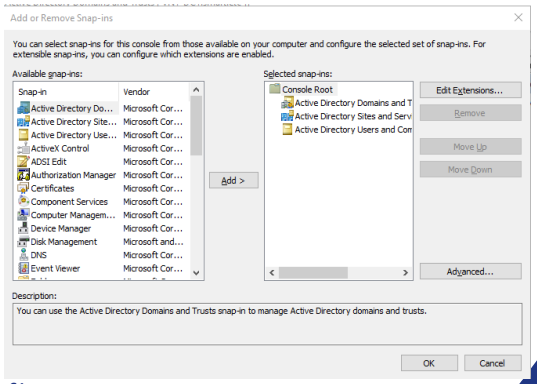
What is MMC - Windows Server | Microsoft Docs

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Creating a custom MMC console

1. Launch mmc.exe
2. On the menu, click **File, Add/Remove Snap-in…**
3. Add the required Snap-Ins.
   Some Snap-Ins may require additional configuration.
4. Save the console using the **File** menu.



[Author an MMC console file: Microsoft Management Console (MMC) | Microsoft Docs]

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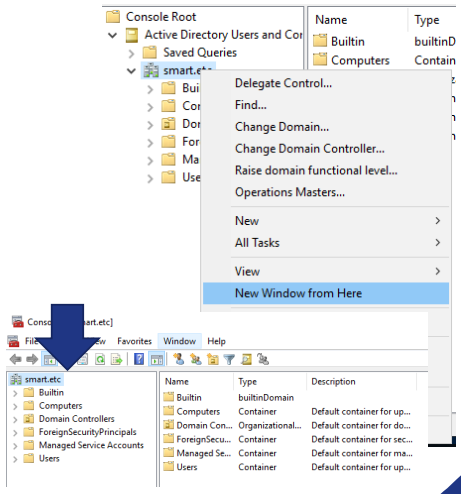
MMC tips & tricks

Too many snap-ins slow down the launch of console

Rename **Console Root**

Focus a console on a single snap-in node, by clicking **New Window from Here** on any node

View, Customize… for additional options



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Practice: Create a custom Microsoft Management Console

On CL1, create a custom Microsoft Management Console with the snap-ins Active Directory Users and Computers and Computer Management and save it to the desktop of all users.

Required time: 5 minutes



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Overview of Server Manager

More modern administrative tool

Graphical tool to install roles and features

Multi-server management

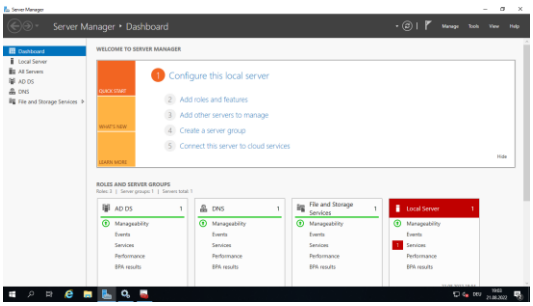
Automatically discovers installed roles and features

Only some modern features can be administered directly in Server manager

Pre-configured MMC consoles and other tools are available on the **Tools** menu

Server Manager | Microsoft Docs

Manage the Local Server and the Server Manager Console | Microsoft Docs



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Server Manager: multi-server management

Adding servers:

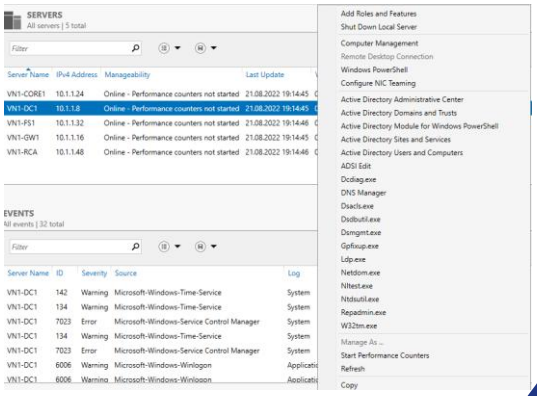
1. On the menu, click **Manage, Add Servers**.
2. On the left pane, roles from remote servers may appear

Managing and removing servers:

1. On the left pane, click **All Servers**
2. Invoke the context menu of any server in the list

   Available commands depend on roles and features installed

Add Servers to Server Manager | Microsoft Docs



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Practice: Explore Server Manager

On CL1, in Server Manager, add all Servers with a name starting with VN or PN and explore Server Manager.

Required time: 5 minutes



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Windows Server administration: Remote Desktop

Special case of local administration

Requires enabling Remote Desktop on the target server (turned off by default)

Local administration has higher security risks

Enabling Remote Desktop imposes even higher security exposure

Typical use case: Running a graphical application setup

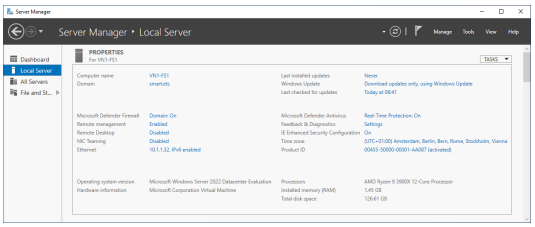
Alternatives

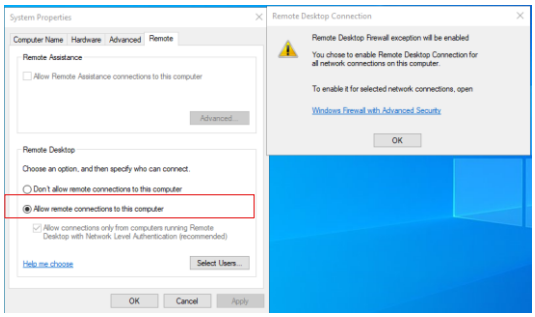
- Remote server administration tools
- PowerShell remoting
- Windows Admin Center

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Windows Server administration using Remote Desktop

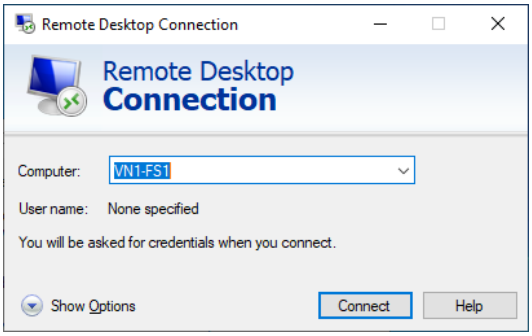


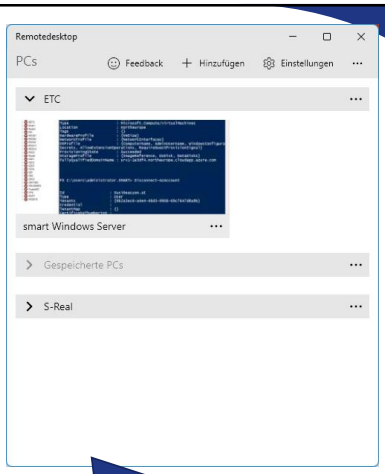


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Using Remote Desktop





Remote Desktop app from the Microsoft store has some advantages including auto-resizing

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Windows Admin Center

Web application management of servers

Agent-less and secure remote management

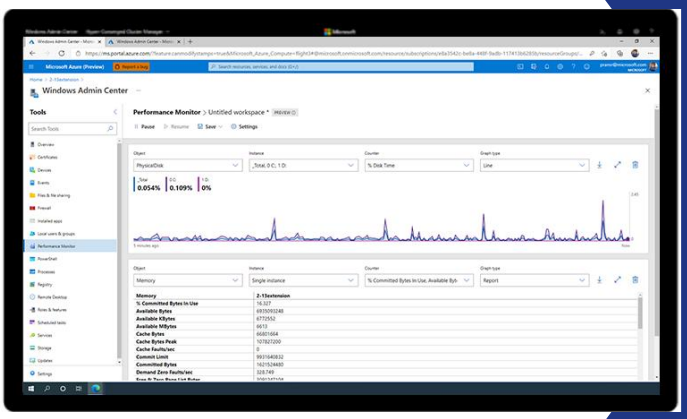
Extensible

Available for local installation or in Azure

Hybrid cloud management

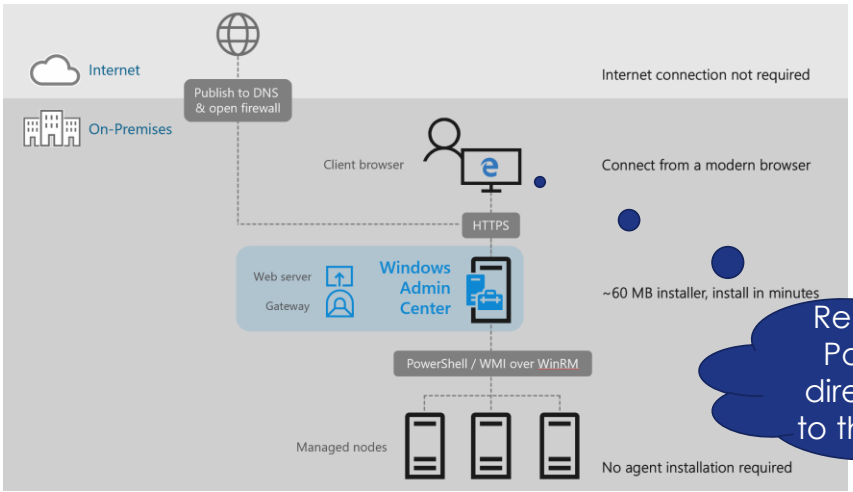
Access to Remote Desktop and remote PowerShell

Windows Admin Center Overview | Microsoft Docs



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Windows Admin Center: How it works

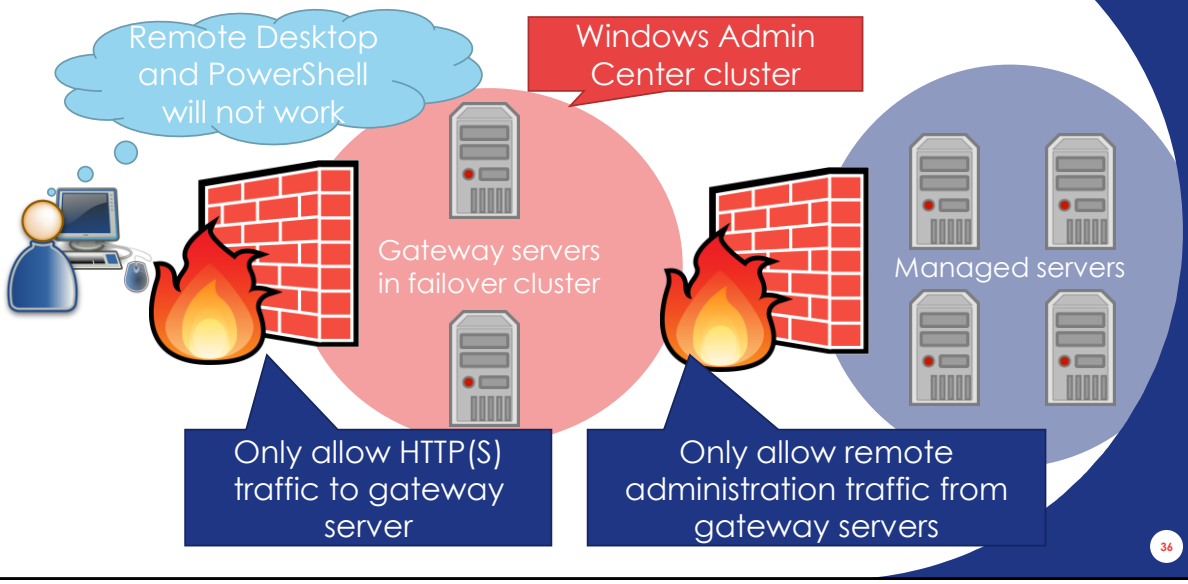


Remote Desktop and PowerShell connect directly from the client to the managed nodes

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Windows Admin Center: Secure architecture



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Practice: Install Windows Admin Center using a script

On VN1-SRV4, to install Windows Admin Center, run **C:\LabResources\Solutions\Install-AdminCenter.ps1**.
Required time: 5 minutes



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Windows Admin Center: managing access

By default, any user that accesses the gateway URL has access and is a Gateway User!

- Admins need to configure users who should be given access

Additional authentication methods

- Require Smartcard logon
- Azure AD logon
  - Allows for MFA, etc.

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Windows Admin Center: managing connections

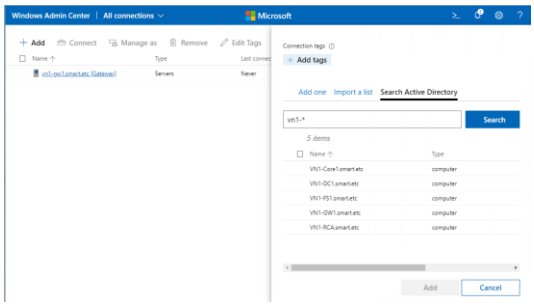
Click on the **Windows Admin Center** banner (top-left)

Add connection

Remove connection

All connections are user-specific by default

Get started with Windows Admin Center | Microsoft Docs



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Windows Admin Center: tags

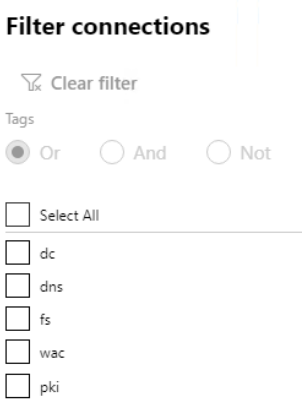
Use tags to, e.g.

- Indicate server roles
- Indicate cost centers

Filter the server list using tags

Get started with Windows Admin Center





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Windows Admin Center: extensions

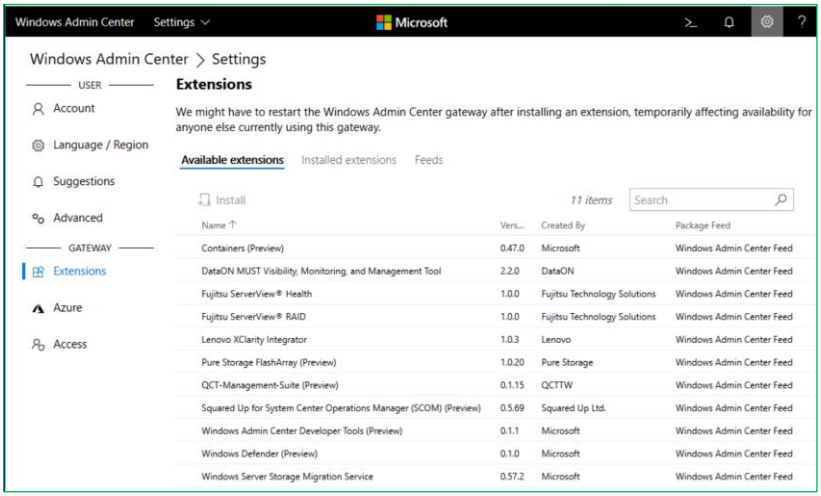
Windows Admin Center is built as an extensible platform
Extension are built using modern web technologies including HTML5, CSS, Angular, TypeScript and jQuery
Documentation available via SDK
Extension can be published to

- Windows Admin Center extension feed (NuGet)
- A different NuGet feed
- A file share

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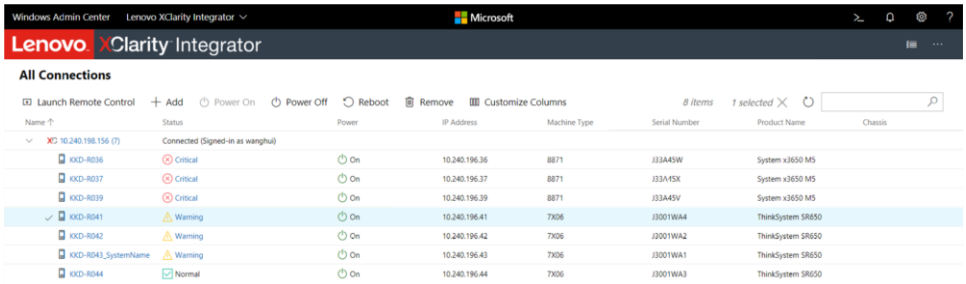
Windows Admin Center: extension examples (1)



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Windows Admin Center: extension examples (2)



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Windows Admin Center: extension examples (3)



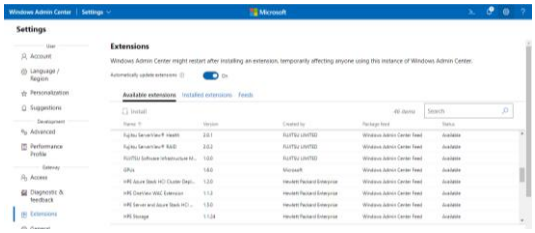
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Windows Admin Center: managing extensions

You must be Gateway Administrator to manage extensions.

1. Click the *Settings* icon on the top-right (gear icon)
2. On the left pane, click **Extensions**

Install and Manage Extensions | Microsoft Docs



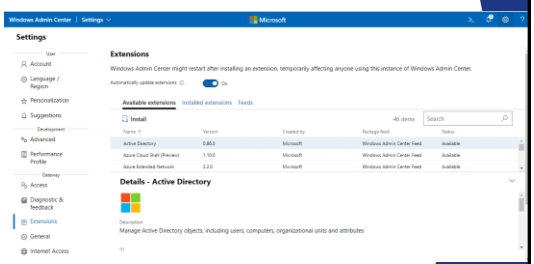
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Windows Admin Center: installing extensions

Installing extensions

1. Click the tab **Available extensions**
2. Select an extension and click **Install**



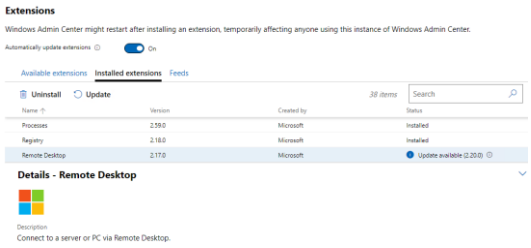
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Windows Admin Center: updating extensions

Updating extensions

1. Click the tab **Installed extensions**
2. Select an extension and click **Update**



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Windows Admin Center: comparison to graphical tools

Pros

- No installation on the client required
- Administered servers can be further isolated
- Administration from non-Windows platforms
- Secure remote administration

Cons

- Additional infrastructure required
  - Can be mitigated by using Windows Admin Center in Azure
- Additional single-point of failure
  - Can be mitigated with a high availability setup
- Not all administration tasks supported
  - Can be mitigated using Remote Desktop or PowerShell from Windows Admin Center in Azure

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Lab: Explore Windows Admin Center

1. Connect
2. Manage connections
3. Explore available administration features
4. Use tags

Required time: 20 minutes



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More information

Server Manager

Configure remote Management in Server Manager | Microsoft Docs

Create and Manage Server Groups | Microsoft Docs

Remote Desktop

Remote Desktop Connection Manager - Windows Sysinternals | Microsoft Docs

Windows Admin Center

Course Windows Server: Fortgeschrittene Administration und Deployment

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Command line administration tools

Command line hosts, Windows Terminal, CMD.EXE, PowerShell



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Why use command line tools?

ETC

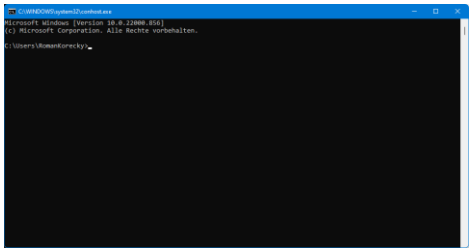
Often faster than graphical tools
More powerful, especially on repetitive tasks
    E.g., reset password of all users of a single department
Reproducible
    Especially beneficial for complex tasks
Can be used as documentation

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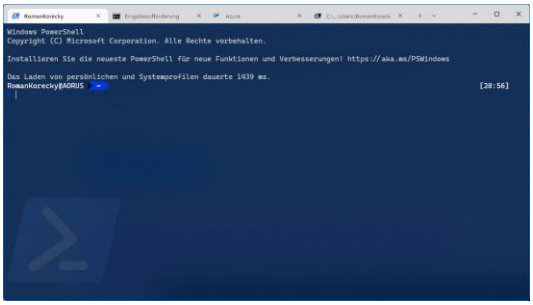
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Command line hosts

CONHOST.EXE



Windows Terminal



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Windows Terminal: Benefits

Broad support of command line interfaces including Azure Cloud Shell, and Windows Subsystem for Linux (WSL)
Customizable color schemes
Customizable keyboard shortcuts
Unicode and UTF-8 character supports (emojis 👍 👌 😊)
GPU accelerated text rendering
Background image support
An overview on Windows Terminal | Microsoft Docs

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Windows Terminal: Installing from GitHub

1. From Releases · microsoft/terminal · GitHub, download the files **Microsoft.Windows.Terminal.*.msixbundle** and **Microsoft.WindowsTerminal.*_PreInstallKit**
2. Extract the file **Microsoft.WindowsTerminal.*_PreInstallKit**
3. Open Windows PowerShell and navigate to the extracted PreinstallKit files.
4. Run
```
Add-AppxPackage `
    Microsoft.UI.Xaml.*_x64__*.appx
```
5. Navigate to the folder containing the file **Microsoft.Windows.Terminal.*.msixbundle**
6. Run
```
Add-AppxPackage `
    Microsoft.Windows.Terminal_*.msixbundle
```

The asterisk stands for additional file name component, that may vary.
For Windows Server, download the Win10 versions.

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Practice: Install Windows Terminal

Install Windows Terminal on VN2-SRV1

Required time: 10 minutes



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Windows Terminal: useful shortcuts

`Ctrl + Shift + P`	Open/close command palette
`Alt + Shift + D`	Duplicate section
`Alt + Shift + Left/Right/Up/Down`	Scale section
`Ctrl + Shift + W`	Close section
`Alt + Shift + Minus`	Split section down
`Alt + Shift + Plus`	Split section right
`Alt + Left/Right/Up/Down`	Move focus
`Ctrl + Shift + N`	New window

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Command line editing

Command **line** is meant literally

A command line can get very long and may wrap

Use arrow keys left/right to navigate within a command line for editing purposes

Do not use up/down for editing the current line!

Use arrow keys up/down to scroll through a history of previous command lines

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CMD.EXE

Traditional command line interpreter of Windows Server

Many commands and concepts are inherited from the infamous command.com, which has its origins in MS-DOS.

Runs internal commands and external commands

External commands may be called *command line tools*.

Commands may accept text input from STDIN and may output to STDOUT

STDIN and STDOUT can be used to pipe command or redirect input or output using symbols like `|`, `<` and `>`

Commands do not share a common syntax

Most commands provide some help using the `/?` switch.

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CMD.EXE: useful commands

`c:` or `d:` ...	Change to the volume
`dir`	List folder contents
`cd`	Change directory (navigate to folders)
`md` or `mkdir`	Make directory (create folder)
`rd` or `rmdir`	Remove directory (delete folder)
`copy`	Copy file(s)
`move`	Move file(s)
`ren`	Rename file
`type`	Type the contents of a file
`del`	Delete file(s)
`xcopy.exe`	Advanced copying of files and folders
`robocopy.exe`	Even more advanced copying of files and folders; fastest file copy tool ☺

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Practice: Explore Windows Terminal

On CL1, update Windows Terminal to the latest version. In Windows Terminal, query the current PowerShell version. In Windows Terminal, in a Command Prompt tab, list the contents of the user profile folder. List the contents of C:\Windows\System32 including sub directories. Explore the shortcuts from the course slide. Explore Windows Terminal settings.
Required time: 10 minutes



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PowerShell

Easy to use
Structured
Fully integrated in OS
Flexible and extensible
Based on .NET Framework
All commands emit objects
Objects are based on classes and have properties and methods
What is PowerShell? - PowerShell | Microsoft Docs

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PowerShell: cmdlets

Verb-Noun

about Command Syntax - PowerShell | Microsoft Docs

There is no space before or after the dash!

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PowerShell: common verbs

New	creates new objects from scratch	`New-LocalUser`
Get	reads objects	`Get-LocalUser`
Set	updates objects	`Set-LocalUser`
Remove	deletes objects	`Remove-LocalUser`
Add	adds an existing object to another existing object Note: There are some infamous modules using Add to create new objects	`Add-LocalGroupMember`

Approved Verbs for PowerShell Commands - PowerShell | Microsoft Docs

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PowerShell: Nouns and modules

Cmdlets are provided by PowerShell modules (and snap-ins)
Some modules are included, some are installed with applications, some can be installed using PowerShell Gallery
Many modules prefix the nouns of their cmdlets
- E.g., the ActiveDirectory module prefixes all nouns with AD
  `Get-ADComputer`

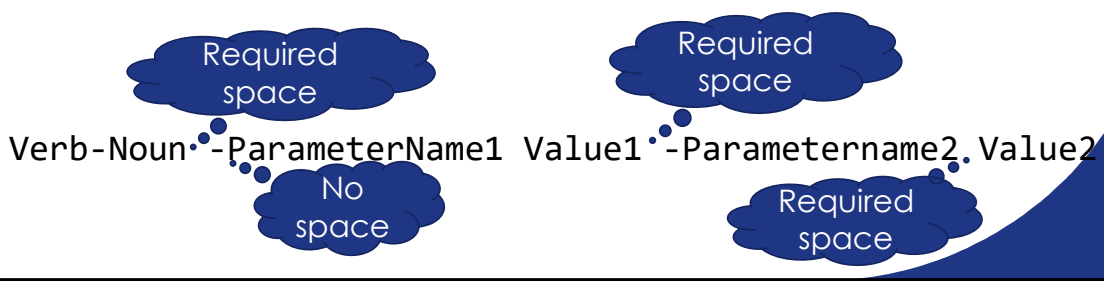
about Modules - PowerShell | Microsoft Docs

65

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PowerShell: parameters

Most cmdlets need additional parameters to run
There are obligatory and optional parameters
PowerShell will ask for obligatory parameters interactively
For some parameters, the parameter name may be omitted
about Command Syntax - PowerShell | Microsoft Docs



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PowerShell: managing modules

Cmdlets are organized in modules

List all available (installed) modules

```
Get-Module -ListAvailable
```

Modules are usually imported automatically, if a command from the module is used. Import a module manually

```
Import-Module Defender
```

Modules are installed together with other software.

Modules can be found and installed, updated and uninstalled from PowerShell Gallery

```
Find-Module *IIS*
Install-Module IISAdministration
Update-Module IISAdminstration
Uninstall-Module IISAdministration
```

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PowerShell: finding commands

```
Get-Command
   Get-Command *user*
   Get-Command -Noun *user*
   Get-Command -Noun *user* -Verb Get
   Get-Command -Module ActiveDirectory
   Get-Command -Noun *user* -Module ActiveDirectory
```

Get-Command (Microsoft.PowerShell.Core) - PowerShell | Microsoft Docs

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PowerShell: getting help

```
Get-Help
   Get-Help Get-Process
   Get-Help Get-Process -ShowWindow
```

The first time you use Get-Help, you might be asked to download the help files. You can do this at any time by running `Update-Help` (administrative rights required).

`about_` topics explain general concepts

```
   Get-Help about_
   Get-Help about_Command_Syntax
```

Get-Help (Microsoft.PowerShell.Core) - PowerShell | Microsoft Docs

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PowerShell: cmdlet equivalents to CMD.EXE commands

CMD.EXE	PowerShell
c: or d: …	Set-Location
dir	Get-ChildItem
cd	Set-Location
md or mkdir	New-Item -Type Directory
rd or rmdir	Remove-Item
copy	Copy-Item
move	Move-Item
ren	Rename-Item
type	Get-Content
del	Remove-Item
xcopy.exe	Copy-Item

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PowerShell: more syntax elements

Single line comment

```
# This is my comment
```

Multi-line comment

```
<#
   It is a good idea to put a lot of comments into your
   scripts. The comments should not explain the code
   itself, but the business requirements leading to the
   code. Comments can also be used to structure the
   code.
#>
```

Wrapping command over multiple lines (backtick)

```
Start-BitsTransfer `
    -Source 'https://aka.ms/WACDownload' `
    -Destination 'C:\LabResources'
```

Notice the backticks and the space before!

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PowerShell: editing tips

Command line editing rules apply

Use TAB key to complete

- Cmdlets
- Parameter names
- Paths (files, registry keys, …)
- Parameter values for validate sets (fixed set of parameter values)



Multiline commands are possible, if PowerShell detects an incomplete line ending with `, |, (, {, …

PowerShell indicates continuing lines with a double prompt >>

Either complete the command or cancel by pressing CTRL + C

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Practice: Find PowerShell commands, get help, and manage modules

On CL1, find a command to create a new local user and display the help for it. Display the help for the command syntax of PowerShell. Install the AzureAD PowerShell module in the Scope of the current user.

Required time: 10 minutes



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PowerShell: pipeline

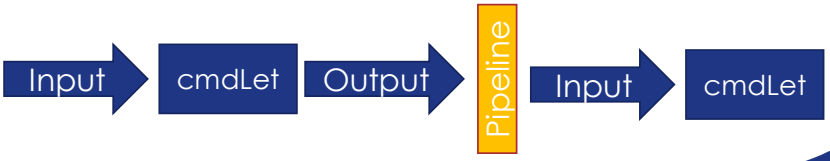
The output objects of one cmdlet are the input objects for following cmdlet.

The result is a very powerful command chain or "pipeline" that is comprised of a series of simple commands.

Use the pipe character | to build pipelines.

```
CmdLet1 | CmdLet2 | CmdLet3
```

about Pipelines - PowerShell | Microsoft Docs



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Classes

Classes are blueprints or templates for objects, defining properties and methods

Objects are instances of classes

Properties store values about the state of an object

Objects of one class have the same properties, but may have different property values

Methods are functions that do something with the object

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PowerShell: accessing properties and methods

Properties and methods can be access using the dot (.) syntax.

```
$someObject.someProperty
```

Methods always have a parameter list in parenthesis, even if it empty

```
$someObject.Method(
    parameterValue1, paramterValue2, …
)
$someObject.Method() # empty parameter list
```

If the dot-syntax should be used directly on a command, the command must be enclosed in parentheses

```
(Get-LocalUser -Name 'Administrator').Description
```

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PowerShell: exploring classes

`Get-Noun | Get-Member`

`Get-Process | Get-Member`

If the output is a list of objects from the same class, Get-Member returns information about a single object.

Use `(Get-Noun).GetType()` to get the real type information about the object.



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Practice: Explore PowerShell return types

On CL1, find cmdlet to receive running processes and installed services. Try the cmdlets and display information about the returned classes.

Required time: 5 minutes



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PowerShell: controlling output

Object-cmdlet can be used with any output.

Typically, they are used within pipelines.

Typical pipeline

Get-Noun | Where-Object | Select-Object | Sort-Object | Format-Table

Projection: Select properties to send to the output

```
Select-Object PropertyName1, Propertyname2, …
```

Selection: Filter objects (see next slide)

```
Where-Object
```

Sorting: Sort objects

```
Sort-Object PropertyName1, PropertyName2, …
```

Formatting: Format output

```
Format-Table
Format-List
```

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PowerShell: Where-Object

```
Where-Object -FilterScript { FilterScript }
```

E. g., `Where-Object { $PSItem.Name –like 'Admin*' }`

`FilterScript` is an expression returning `$true` or `$false`.

`FilterScript` must be written in braces, because it is a script.

In a `FilterScript`, use `$PSItem` (or `$_`) to reference the object of consideration.

`$PSItem` is typically followed by a dot (.) and the name of the property under consideration.

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What is $PSItem?

`$PSItem` is an automatic variable containing the current object in the pipeline object
`$_` is an alias of `$PSItem`
Common uses:
- In the script block in `ForEach-Object`
- In the script block of `Where-Object`

about PSItem - PowerShell | Microsoft Learn

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PowerShell: operators

`-eq`	Equal
`-ne`	Not equal
`-gt`	Greater than
`-ge`	Greater or equal
`-lt`	Less than
`-le`	Less or equal
`-like`	Wildcard comparison
`-not`	Not
`-and`	And
`-or`	Or

PowerShell supports typical calculations like:
3 + 5
3 * 5
3 - 5
3 / 5

```
Get-Help about_Operators
Get-Help about_Arithmetic_Operators
Get-Help about_Comparison_Operators
Get-Help about_Logical_Operators
Get-Help about_Type_Operators
```

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Practice: Manage PowerShell output

On CL1, display all properties of the Windows Terminal process, start all services, configured to start automatically, but are not currently running.

Required time: 10 minutes



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PowerShell: variables

Benefits

- Save values, needed by multiple commands
- Saves the output from commands
- Saves you from typos
- Can be completed using TAB
- Assign once, use many times

Usage

- Variables start with a $ sign
- Assigning variables

```
$username = 'Max'
$processes = Get-Process
```

- Check the content of a variable, by simply executing the variable name

```
$username
```

- Use the variable name instead of a value

```
New-LocalUser `
    -Name $username
```

about Variables - PowerShell | Microsoft Docs

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PowerShell: string literals

Strings in single quotes (`'string'`) are literals.
Strings in double quotes (`"string"`) may contain variables or other expressions.
Use $() for complex expressions in double-quoted strings.

```
$firstname = 'Max'
$lastname = 'Mustermann'
$displayname = "$firstname $lastname"
$displayname2 = '$firstname $lastname'
$username = "$($firstname[0]).$lastname“
```

about Quoting Rules - PowerShell | Microsoft Learn

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Practice: Work with PowerShell variables

Store 'cl1' and 'ad.adatum.com' in separate variables and create a third variable for a FQDN using the two variables. Calculate the time until December 25th of the current year. Calculate the number of bytes of 7 TB.
Calculate the time until December 25th of the current year.
Calculate the number of bytes of 7 TB.
Required time: 5 minutes



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PowerShell: arrays

Many cmdlets return a list of objects, called arrays

Build your own arrays using the @() literal

```
$servers = @('VN1-DC1', 'VN1-GW1', 'VN1-FS1', 'VN1-Core1')
```

Special operators

```
$servers -contains 'VN1-FS1'
$servers -notcontains 'VN1-GW1'
'VN1-DC1' -in $servers
'VN1-Core1' -notin $servers
```

The values or objects of an array can be addressed using the bracket syntax (also applies to strings)

```
$servers[0]
$servers[0,2]
$servers[0..2]
```

Values of a single property of all array items

```
$processes = Get-Process
$processes.Name
```

about Arrays - PowerShell | Microsoft Docs

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PowerShell: working with items

Array as input object in a pipeline

```
$processes = Get-Process `
    -Name msedge
$processes | Stop-Process
```

Why does this work?

```
Get-Help Stop-Process
```

```
Stop-Process [-InputObject] <System.Diagnostics.Process[]>
[<CommonParameters>]
```

Notice the brackets behind the type name

Iterate through array

```
$userRecords = Import-Csv -Path $path
$userRecords | ForEach-Object –ScriptBlock
{
    New-ADUser `
        -Name $PSItem.Name `
        -DisplayName $PSItem.Name `
        -GivenName $PSItem.Firstname `
        -Surname $PSItem.Lastname `
        -Username $PSItem.UPN `
        -SamAccountName $PSItem.Username
}
```

`$PSItem` (or `$_`) references one object of the array on each iteration.

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Practice: Work with PowerShell arrays

On CL1, query the running services on VN1-SRV1, VN1-SRV2, VN1-SRV4, VN1-SRV5, VN1-SRV10 with a single command using iteration.

Required time: 10 minutes



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PowerShell: providers and drives

PowerShell drives facilitate access to hierarchical structures

Drives are provided by PowerShell Providers

Most common providers

- FileSystem
- Registry

Most common roots

- C, D, E, …
- HKCU
- HKLM

```
$path = `
'HKCU:\Software\Microsoft\Windows\CurrentVersion'
Set-Location $path
Get-ChildItem
$keyName = 'TestKey'
New-Item $keyName
$keyPath = "$path\$keyName"
$propertyName = 'TestProperty'
Set-ItemProperty `
    -Path $keyPath -Name $propertyName `
    -Value 'My custom value' -Type String
Remove-ItemProperty `
    -Path $keyPath -Name $propertyName
Remove-Item $keyPath
```

about Providers - PowerShell | Microsoft Docs

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Practice: Registry editing using PowerShell

On CL1, list the keys and values of the registry key 'HKLM:\system\CurrentControlSet\services\Tcpip\Parameters'.

Within that key, set the value IPEnableRouter to 1 and verify the change.

Remove the value IPEnablerouter and verify the change again.

Required time: 5 minutes



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PowerShell: functions and aliases

Aliases are other names for cmdlets or Functions

```
Get-Alias
```

Many cmdlets are implemented as functions

```
Get-Command –Type Function
Get-Command -Type Function |
Where-Object {
    $PSItem.Source -eq ''
}
```

about Aliases - PowerShell | Microsoft Docs

Pre-defined functions for well-known CMD.EXE or bash commands

`A:/B:/C:/…/Z:`	Sets the location to the volume
`cd..`	Sets the location to the upper forlder
`cd\`	Sets the location to the root folder
`help`	Calls `Get-Help | more`
`mkdir`	Calls `New-Item –Type Directory`
`more`	Calls `Get-Content` with `more.exe`
`Pause`	Calls `Read-Host`

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PowerShell: aliases

Shortcuts to CmdLets or functions

?	Where-Object
fl	Format-List
foreach	ForEach-Object
ft	Format-Table
gm	Get-Member
kill	Stop-Process
select	Select-Object
sort	Sort-Object
where	Where-Object

Aliases for well known CMD.exe or bash commands

%	ForEach-Object
cd	Set-Location
copy/cp	Copy-Item
del/rm	Remove-Item
dir/ls	Get-ChildItem
kill	Stop-Process
man	help
md	mkdir
mv	Move-Item
rd/rm/rmdir	Remove-Item
ren	Rename-Item

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PowerShell: remoting

Uses WS-CIM (protocol of Web Based Enterprise Management standard)

Enable remote administration (enabled by default on Windows Servers)

```
Enable-PSRemoting
```

Start an interactive session to a remote computer

```
Enter-PSSession $computername
```

Close an interactive remote session

```
Exit-PSSession
```

about Remote - PowerShell | Microsoft Docs

about Using - PowerShell | Microsoft Docs

WBEM | DMTF

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PowerShell remoting scripts

Run a command or a short script on a remote computer

`$using` maps, a variable from the local session into the remote session

```
$processName = 'sol'
Invoke-Command -ComputerName $computername -ScriptBlock
{
    # These commands will run on the remote computer
    Get-Process –Name $using:processName
}
```

A long ScriptBlock might make troubleshooting difficult.

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PowerShell remoting sessions

Create a session

```
$session = New-PSSession –ComputerName $computerName
```

Run a script in a session

```
Invoke-Command -Session $session -ScriptBlock {
    # Some commands
}
```

Remove a session

```
Remove-PSSession -Session $session
```

Connect or reconnect to a session interactively

```
Connect-PSSession –Session $session
```

Exit from a session without closing it

```
Exit-PSSession
```

Using sessions, you can run shorter ScriptBlocks without performance drawback.

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PowerShell remoting alternatives

Some cmdlets support the parameter `-ComputerName`

- Cmdlets supporting `–ComputerName` use various TCP ports and protocol for remote connections
- Usually these are the same, the MMC snap-ins use

Common Information Model (CIM) Session

- CIM is a standard to represent management information

  CIM | DMTF
- Uses WS-CIM (same as PowerShell remoting)
- Direct query to CIM instead of embedding PowerShell statements
- Usually faster than PowerShell remoting

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Using CIM sessions

Creating a CIM session

```
$cimSession = New-CimSession -ComputerName $computerName
```

Using a CIM session

- Many cmdlets have a `-CIMSession` parameter
- Provide the CIM session as value to the `–CIMSession` parameter

```
Set-NetFirewallRule `
    -CimSession $cimSession `
    -Name ComPlusRemoteAdministration-DCOM-In `
    -Enabled True `
    -Profile Domain
```

Removing a CIM session

```
Remove-CimSession $cimSession
```

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Practice: PowerShell remoting

On CL1, check the value IPEnableRouter on VN1-SRV2 using an interactive remote session. Check the same value on VN1-SRV5 by invoking remote commands.
Required time: 5 minutes



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More information

Windows Terminal
Windows Terminal tips and tricks | Microsoft Docs
Windows Terminal Cascadia Code | Microsoft Docs
Windows Terminal Custom Prompt Setup | Microsoft Docs
Windows Terminal SSH | Microsoft Docs
Windows Terminal Tab Title Setup | Microsoft Docs
PowerShell
Course SWAT-PS - Automating Administration with PowerShell
Visual Studio Code - Code Editing. Redefined

100

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Common administrative tasks

Managing roles and features, managing services, reviewing events, configuring firewall



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Windows Server roles and features

Roles are services, a server can provide on the network

- Examples: Domain Services, File Services, Remote Desktop Services, Web Server, …

Features are functionalities supporting the roles, administrators or availability, usually not providing a service on the network

- Examples: Failover Clustering, Group Policy Management, Network Load Balancing, Windows Server Backup

Role services are sub-services of a role that can be installed separately

Installing a role or feature can

- Install binaries
- Install additional dependencies (roles or features)
- Add services
- Configure the Windows Defender firewall

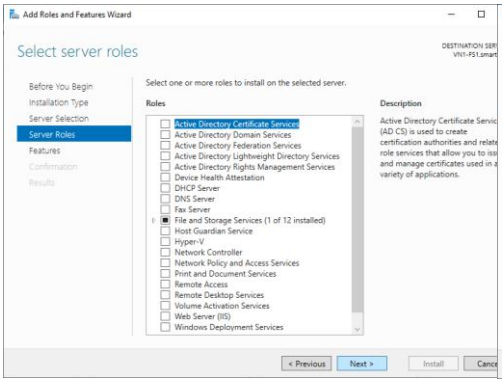
Differentiation between roles and features is artificially. In PowerShell, everything is a feature.

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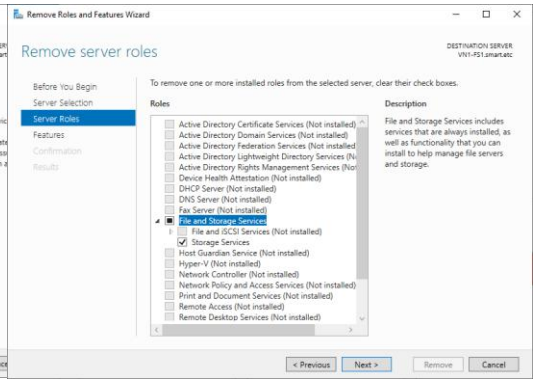
102

Managing Windows role and features using Server manager

Manage, Add Roles and Features



Manage, Remove Roles and Features



Install or Uninstall Roles, Role Services, or Features | Microsoft Docs

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Practice: Install roles using Server Manager

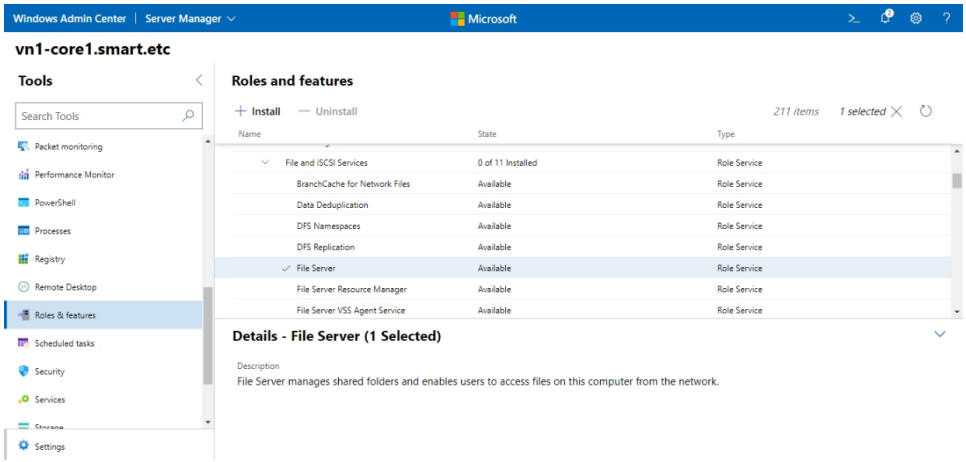
On CL1, use Server Manager to install the File Server role on VN1-SRV10.

Required time: 5 minutes



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Managing roles and features using Windows Admin Center



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Practice: Install roles using Windows Admin Center

On CL1, use Windows Admin Center to install the Certification Authority Web Enrollment role service on VN1-SRV2.

Required time: 10 minutes



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Managing features using PowerShell

List Windows Roles/Features and their status

`Get-WindowsFeature`

Install Windows Features

`Install-WindowsFeature`

`Add-WindowsFeature # Alias for Install-WindowsFeature`

Uninstall Windows Features

`Uninstall-WindowsFeature`

`Remove-WindowsFeature # Alias for Uninstall-WindowsFeature`

Remember:
In PowerShell,
roles are features

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Practice: Manage features using PowerShell

On CL1, use PowerShell to install the Windows Server Backup feature on VN1-SRV1, VN1-SRV2, VN1-SRV4, VN1-SRV5, and VN1-SRV10.

Required time: 10 minutes



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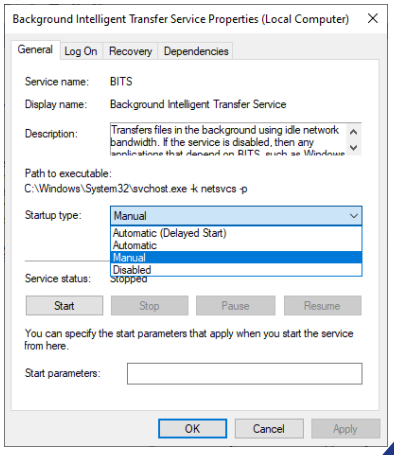
108

Services

Run in the background without a user being logged in

Startup Types

- Disabled
- Manual
- Automatic (Delayed Start)
  - Starts 1 – 2 minutes after the computer has completed the start procedure
- Automatic



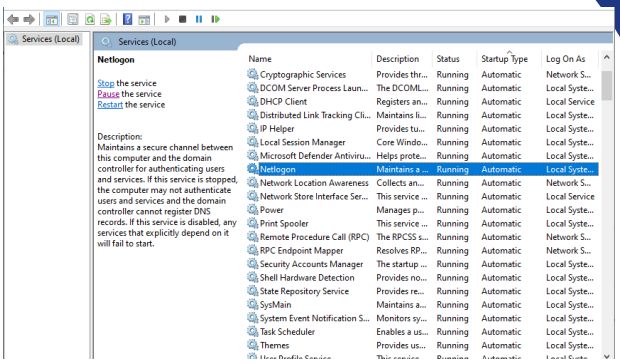
Managing Services using MMC

Edit properties of a service

Start, Stop, Restart a service

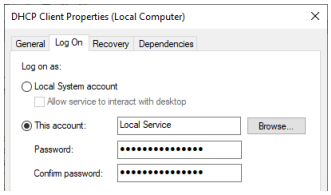
Some services support Pause/Resume

- Effect of Pause depends on service
- In general, pausing a service stops it from accepting new connections, but still serves existing connections



Services: Security context

Service account	Who manages the password	Computer independent (e.g. for clusters)	Local rights	Network service access
Local System	Computer (automatic)	No	Administrator	Yes
Local Service	Computer (automatic)	No	User	No
Network Service	Computer (automatic)	No	User	Yes
Group managed service account	Domain controller (automatic)	Yes	User	Yes
User account	Administrator (manual)	Yes	User	Yes

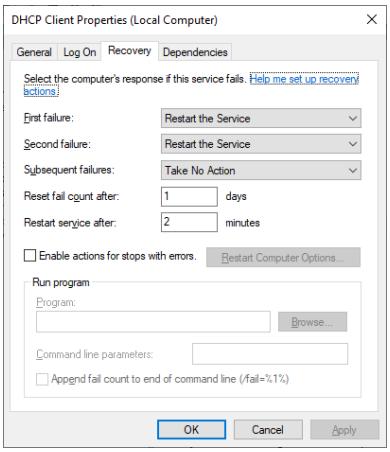


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Services: Recovery

Use for troubleshooting purposes

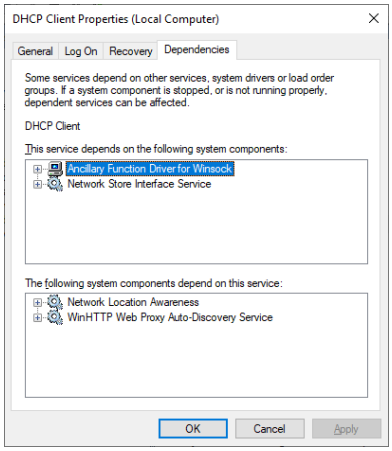


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Services: Dependencies

Services may depend on other services
Other services may depend on a service
Services with a startup type of Manual may be started by dependent services with a startup type of Automatic
Disabling services may cause other services to fail

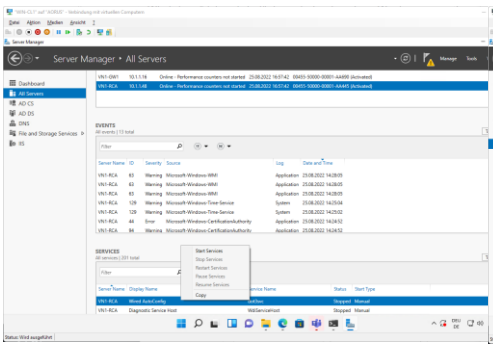


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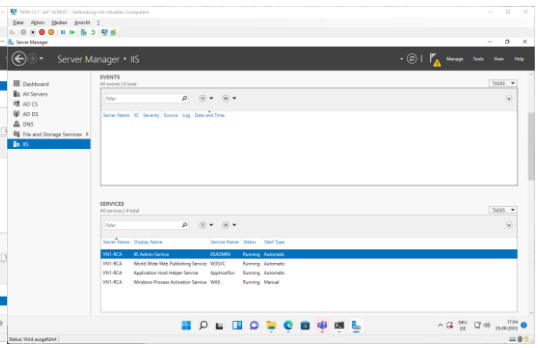
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Managing Services using Server Manager

All services on a server



Role specific service on a server

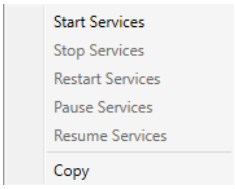


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Managing Services using Server Manager: Features

Multi-Service
- Start
- Stop
- Restart
- Pause
- Resume

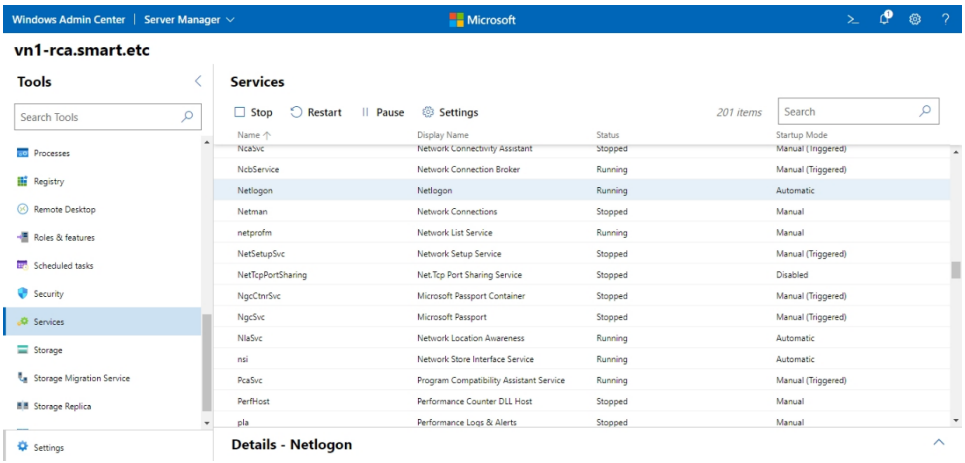
Does not allow to edit service properties



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Manage Services using Windows Admin Center

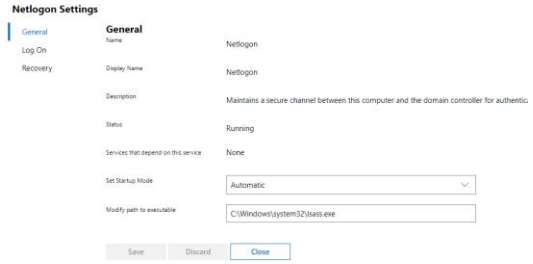


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Manage Services using Windows Admin Center

Most of the features of MMC
Startup type **Automatic (Delayed Start)** is missing



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Practice: Manage Services using Windows Admin Center

On CL1, stop the W32Time service on VN1-SRV5 and disable it. Restart VN1-SRV5 and verify the changes. Then set W32Time to start automatically and start it again.
Required time: 5 minutes



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Manage Services using PowerShell

List service(s) and their status	`Get-Service`
Change service properties	`Set-Service`
Start service	`Start-Service`
Stop service	`Stop-Service`
Restart service	`Restart-Service`
Pause service	`Suspend-Service`
Resume service	`Resume-Service`

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Practice: Manage services using PowerShell

On CL1, list the services on VN1-SRV1 with their status, name, display name, and start type.

Stop the W32Time service on VN1-SRV1, VN1-SRV2, and VN1-SRV10 and disable it.

Restart VN1-SRV1, VN1-SRV2, and VN1-SRV10 and verify the changes.

Then set W32Time to start automatically and start it again.

Required time: 10 minutes



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Events

Messages from the operating system or from services not intended for the service consumer, are called events and written to the Windows log system

Many applications provide additional information in their own log files (usually text files)

Windows Events - Win32 apps | Microsoft Docs

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Event logs and levels

Traditional event logs
- **System** for operating system events
- **Security**
- **Application** for application and services events

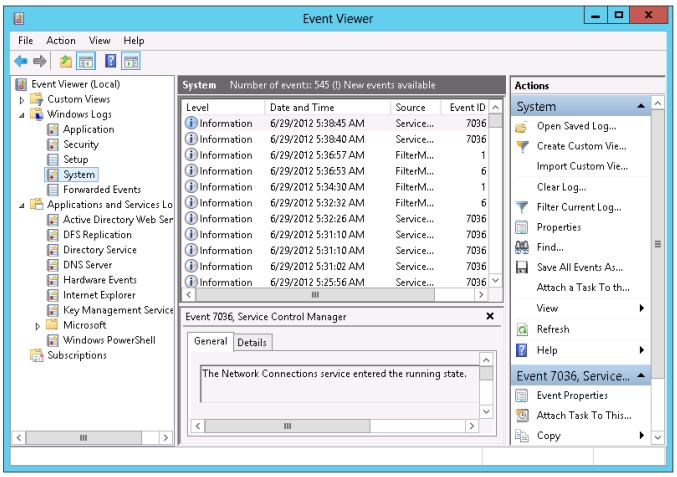
Additional event logs
- **Setup** for feature and update installation events

Application and Service Logs

Event levels
- Critical
- Error
- Warning
- Information
- Verbose

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Viewing events using Event Viewer

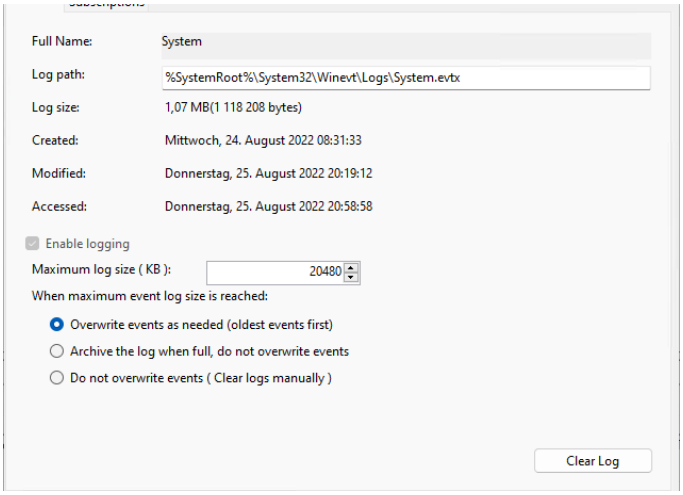


123

Configuring event logs

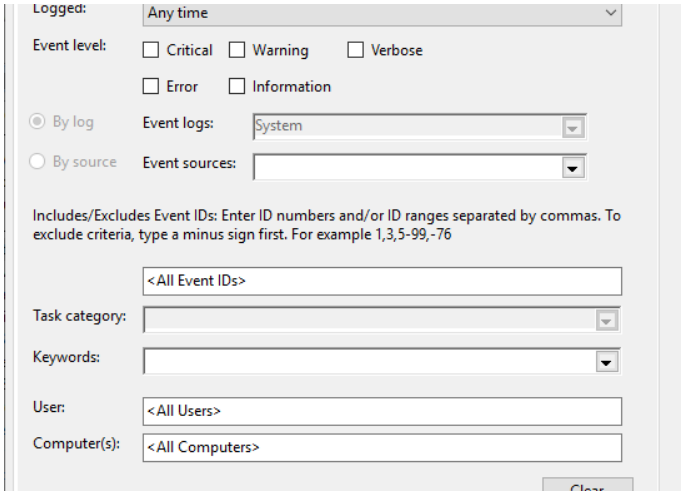
Logging cannot be disabled on default logs like **System**, **Application**, and **Security**.

The option **Archive the log when full, do not overwrite events** archives the log files in the Log path from above.



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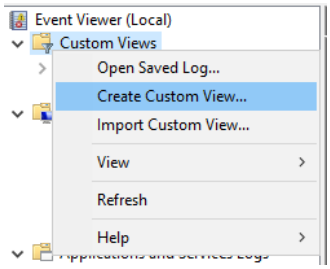
Filtering events

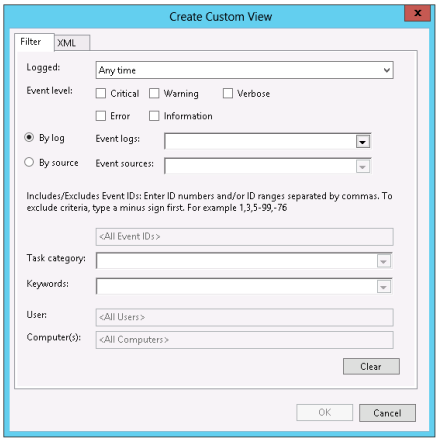


Context menu of the log, **Filter Current Log…**

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Custom event views





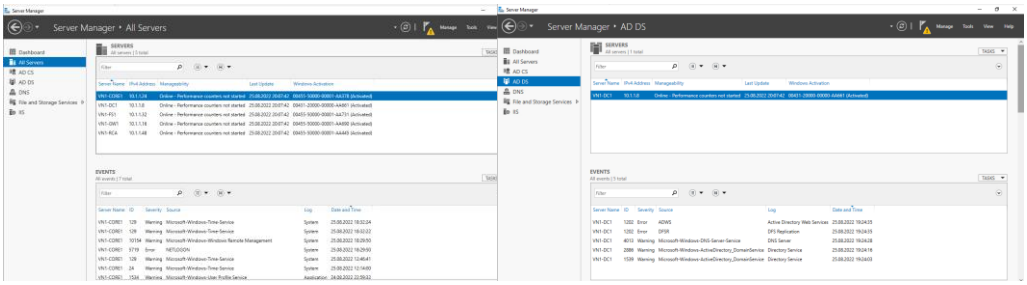
126

126

Viewing events in Server Manager

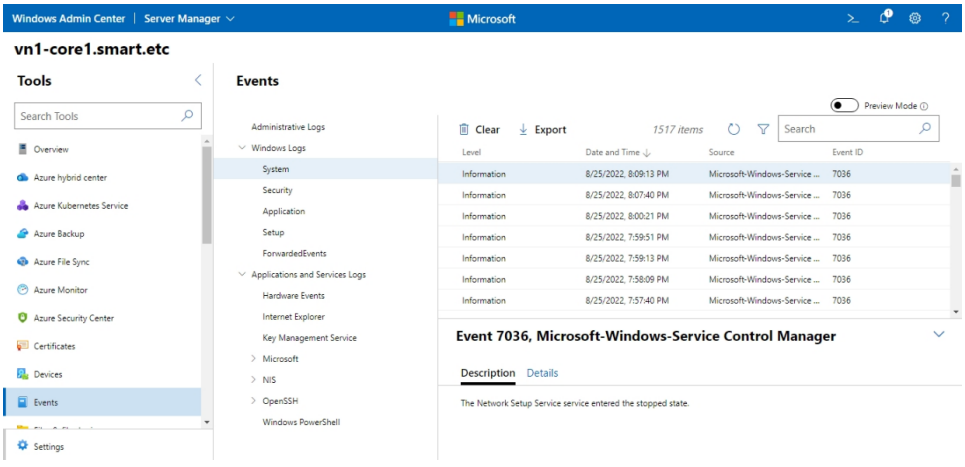
By server

By role



127

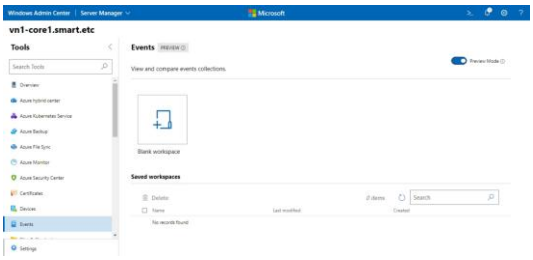
Viewing Events in Windows Admin Center

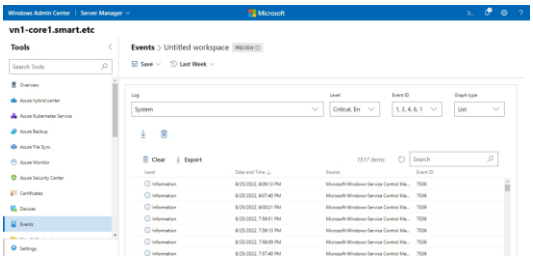


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Viewing Events in Windows Admin Center (Preview)





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Practice: View events in Windows Admin Center

On CL1, view the events from the System log on VN1-SRV1 using Windows Admin Center. In the preview version, create a workspace that includes all warning, error, and critical events from the System and Application logs.

Required time: 5 minutes



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Viewing events using PowerShell

View events from classic event logs such as Application, System, or Security	`Get-EventLog`
Configure classic event log	`Set-Eventlog`
List all logs	`Get-WinEvent -ListLog *`

Configure a log	E.g.
	`$log = Get-WinEvent -ListLog $logName`
	`$log.MaximumSizeInBytes = 1GB`
	`$log.SaveChange()`

Get events from a log	`Get-WinEvent –LogName $logname`

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Practice: View events using PowerShell

On CL1, list the latest 20 events from the system log on VN1-SRV1.

List all warnings, errors and audit failures from the system log on VM1-SRV1.

List all details for the oldest the events that DSC generated on VN1-SRV1.

Required time: 10 minutes



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Windows Defender Firewall

Local Firewall for IPv4/IPv6 traffic
Filters inbound and outbound traffic
Manages connection security (IPSec)
Can be managed directly or through Group Policy
Windows Defender Firewall with Advanced Security (Windows) - Windows security | Microsoft Docs

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Windows Defender Firewall connection profiles

Connection profiles are network adapter specific
Different connection profiles can apply to different adapters on the same computes

Available connection profiles

- Domain
  Automatically applied on domain members, if connection to domain controller is possible
- Private
  Protected networks like intranets
- Public
  Untrusted networks like the Internet or public WLANs
  Most restrictive profile

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Managing Windows Defender Firewall connection profiles

View connection profiles

```
Get-NetConnectionProfile
```

Change connection profile

```
Set-NetConnectionProfile `
    -InterfaceAlias $interfaceAlias `
    -NetworkCategory Private
Set-NetConnectionProfile `
    -InterfaceAlias $interfaceAlias `
    -NetworkCategory Public
```

Instead of `InterfaceAlias`, `InterfaceIndex` can be used.

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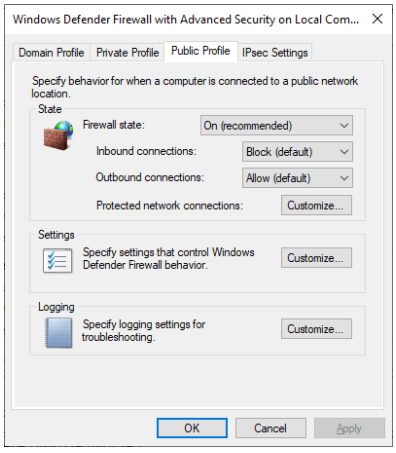
Windows Defender Firewall properties: MMC

For each profile

Firewall state: on/off

Inbound and outbound connection

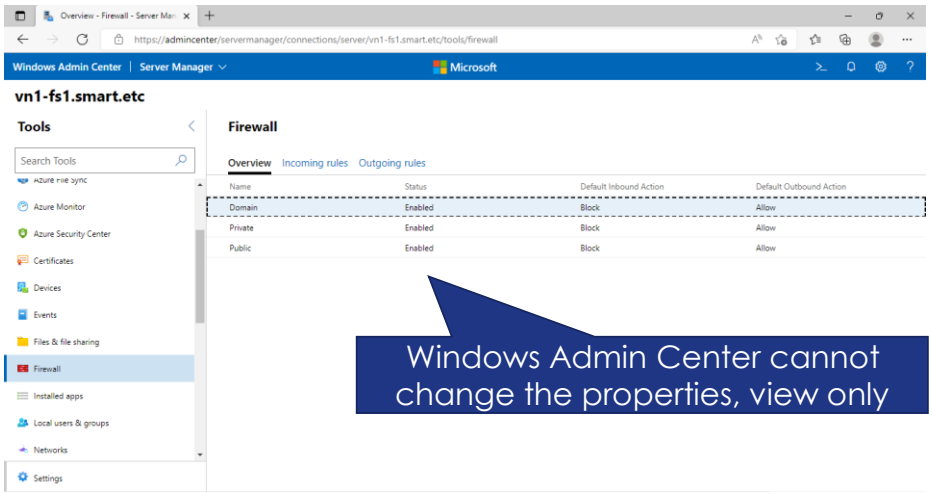
- Block
  Default for inbound connections
- Block all connections
  Disables all allow rules
- Allow
  Default for outbound connections



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Windows Defender Firewall properties: Windows Admin Center



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Windows Defender Firewall properties: PowerShell

Get the properties

```
Get-NetFirewallProperties
```

Enable or Disable the firewall for a connection profile

```
Set-NetFirewallProfile -Name Domain, Public, Private -
Enabled False
Set-NetFirewallProfile -Name Domain, Public, Private -
Enabled True
```

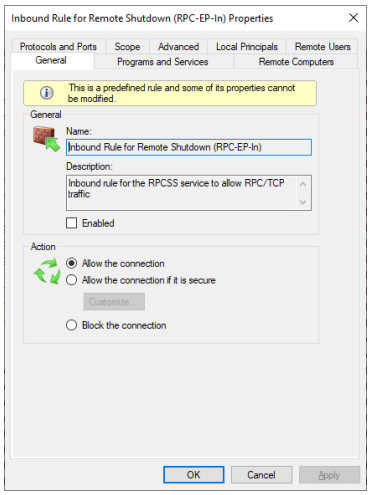
Configure default action for incoming/outgoing traffic

```
Set-NetFirewallProfile `
    -Name Domain, Public, Private `
    -DefaultInboundAction Block
Set-NetFirewallProfile `
    -Name Domain, Public, Private `
    -DefaultOutboundAction Allow
```

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Managing Windows Defender Firewall rules: MMC General



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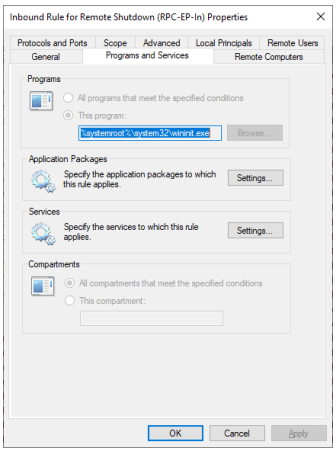
139

Managing Windows Defender Firewall rules: MMC Programs and Services

Programs
- Applies to all connections to/from an executable

Application Packages
- Applies to all connections to/from an application package (appx)

Services
- Applies to all connections to/from a service



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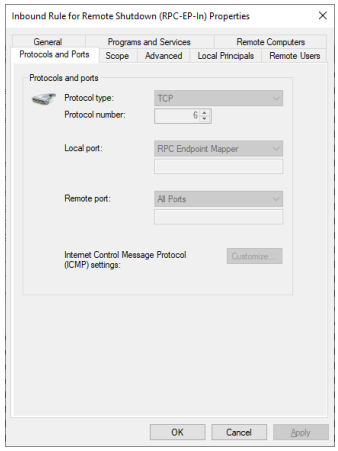
Managing Windows Defender Firewall rules: MMC Protocols and Ports

Apply rule to a certain protocol

Local and remote port available for TCP/UDP only

Multiple ports can be specified as lists or ranges

E.g. 80, 443, 8080 - 8089



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Managing Windows Defender Firewall rules: MMC Scope

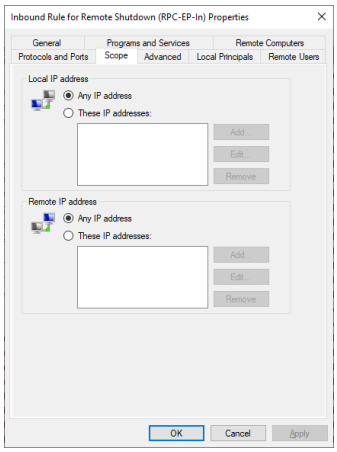
Single IP address

Subnets in CIDR notation

E.g. 192.168.1.0/24

Predefined set of computers

- Local subnet
- Intranet
- Remote Corp Network
- Internet
- …



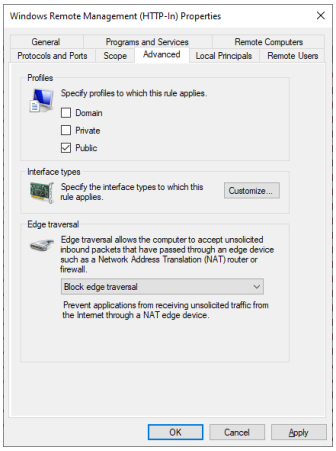
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Managing Windows Defender Firewall rules: MMC Advanced

Apply rule to a connection profile
- Domain
- Private
- Public

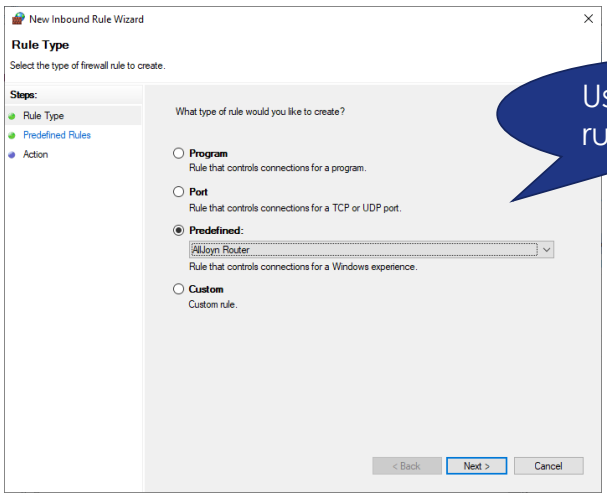
Apply rule to an interface type
- Local area network
- Remote access
- Wireless



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Creating Windows Defender Firewall rules in MMC



Use predefined rules, if possible

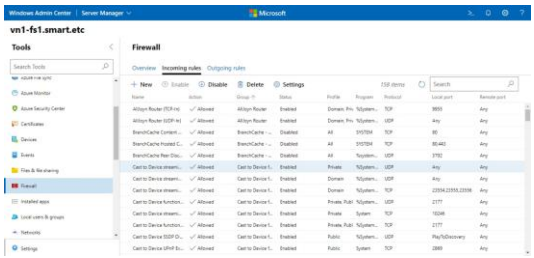
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Managing Windows Defender Firewall rules using Windows Admin Center

Limited rule property support

- Name
- Description
- Direction
- Action
- Enabled/Disabled
- Protocol
- Local/Remote port
- ICMP types
- Profiles



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Managing Windows Defender Firewall rules using PowerShell

Get firewall rules

```
Get-NetFirewallRule
```

Enable a firewall rule

```
Set-NetFirewallRule `
    -Name $name `
    -Enabled True
```

Disable a firewall rule

```
Set-NetFirewallRule `
    -Name $name `
    -Enabled False
```

Enable a rule group

```
Get-NetFirewallRule `
    -DisplayGroup `
        $displaygroup |
Set-NetFirewallRule `
    -Enabled True
```

Create a firewall rule

```
New-NetFirewallRule
```

Remove a firewall rule

```
Remove-NetFirewallRule –Name $name
```

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Lab: Manage servers remotely using Microsoft Management Console

1. Evaluate remote administration using Microsoft Management Console and PowerShell
2. Configure Windows Defender Firewall rules for remote administration
3. Verify the ability of remote administration using Computer Management

Required time: 15 minutes



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Manual setup and first configuration

Setup, Desktop Experience, Core Experience, SConfig



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Practice: Install Windows Server with Desktop Experience manually

Run a script to create a new VM and install Windows Server Datacenter Edition with Desktop Experience on it.
Required time: 10 minutes



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Post-installation configuration settings

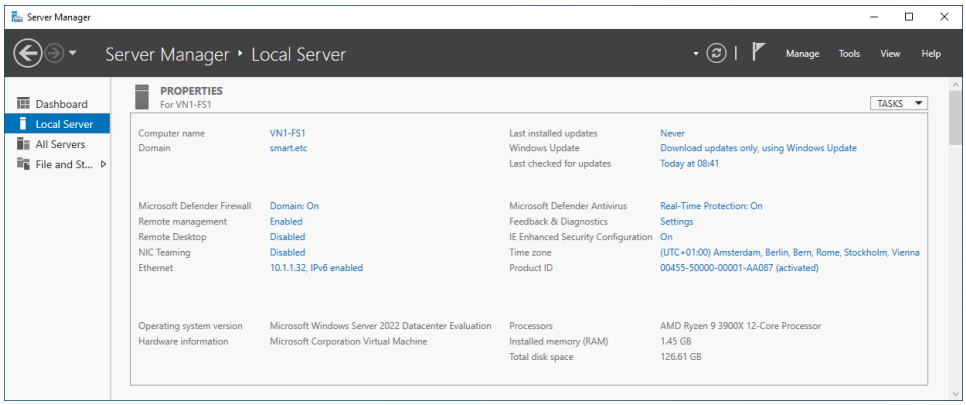
After you have installed Windows Server, you must complete the following:

- Configure the IP address
- Set the computer name
- Join an Active Directory domain
- Configure the time zone
- Enable automatic updates
- Add roles and features
- Enable Remote Desktop
- Configure Windows Firewall settings

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Post-Installation Configuration in Server Manager



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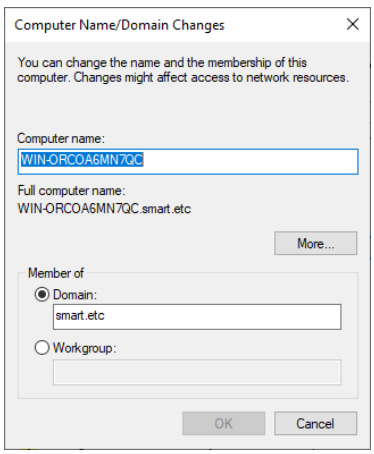
151

How to Join the Domain

Information necessary for a domain join:

- Domain name
- Account with permission to join computer to domain

Note: Every user is allowed to join 10 Computers to the domain (configurable)



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Server Core

A more security-enhanced, less resource-intensive installation option

Default and recommended installation option for Windows Server

Managed locally by using sconfig.cmd, cmd.exe and Windows PowerShell

With remote management enabled, you rarely need to sign in locally

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Server Core vs Server with Desktop Experience install options

Component	Server Core	Server with Desktop Experience
User interface	Minimal, command line driven	Standard Windows GUI
Disk space	Small	Large
Intall configure, uninstall server roles locally	PowerShell	Server Manager or PowerShell
Roles and Features	Some roles and features not available Some features can be installed with FOD	All roles and features available
Remote management	Yes	Yes
Potential attack surface	Reduced	Full
Microsoft Management Console	Optional	Installed

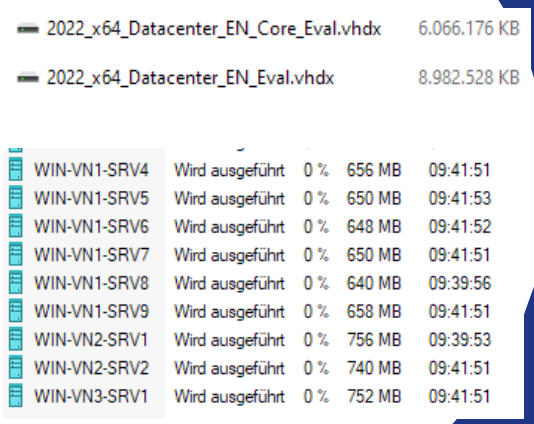
Server Core vs Server with Desktop Experience install options | Microsoft D

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Server Core in numbers

Base image almost 3 GB smaller
About 100 MB less memory consumption



All servers with identical base configuration without any roles installed. WIN-VN2-* and WIN-VN3-SRV2 have Desktop Experience installed.

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Practice: Install Windows Server manually

Run a script to create a new VM and install Windows Server Datacenter Edition on it.

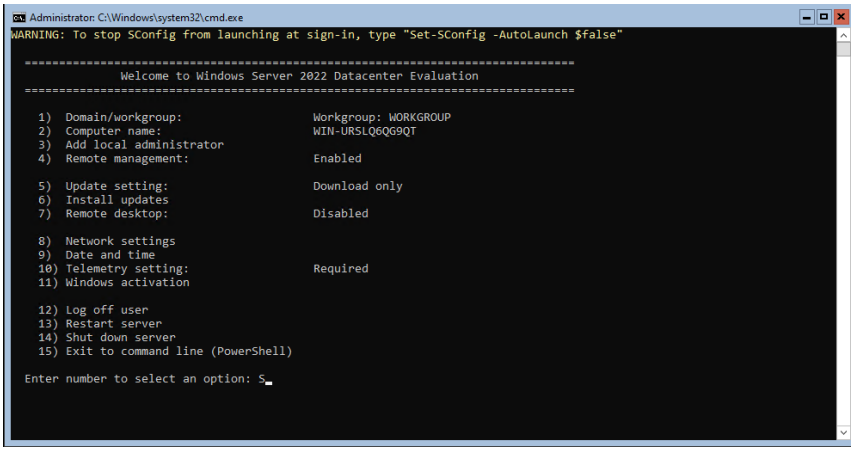
Required time: 10 minutes



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Configuring Windows Server using Sconfig



Configure a Server Core installation of Windows Server and Azure Stack HCI with the Server Configuration tool (SConfig) | Microsoft Docs

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Server Core App Compatibility Feature on Demand (FOD)

Optional feature package for Server Core installations

- Includes a subset of binaries and packages from Server with Desktop Experience
- Increases compatibility with applications

Server Core App Compatibility Feature on Demand (FOD) in Windows Server | Microsoft Docs

Features that can be added:

- Microsoft Management Console
  - Event Viewer
  - Device Manager
  - Failover Cluster Manager
- Performance Monitor
- Resource Monitor
- File Explorer
- Windows PowerShell
- Optional: Internet Explorer

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Downloading and mounting the Windows Server Languages and Optional Features ISO image file

1. Download the Windows Server Languages and Optional Features ISO image file from Volume Licensing Service Center, Microsoft Evaluation Center, or Visual Studio portal.
2. Mount the image file

```
# Put the path to the FOD.iso between the quotes
$isoPath = ''
$fodIso = Mount-DiskImage –ImagePath $isoPath
$fodDriveLetter = ($fodIso | Get-Volume).DriveLetter
```

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Installing the App Compatibility FOD

1. Installing the Server Core App Compatibility

```
$name = 'ServerCore.AppCompatibility~~~~0.0.1.0'
```

- With internet access

```
Add-WindowsCapability -Online -Name $name
```

- Without internet access

```
$source =
"${fodDriveLetter}:\LanguagesAndOptionalFeatures\"
Add-WindowsCapability -Online -Name $name -Source
$source -LimitAccess
```

2. Restart Server

```
Restart-Computer
```

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Installing Internet Explorer 11 (Optional)

Requires Server Languages and Optional Features ISO image file!

```
$packagePath = `
"${fodDriveLetter}:\LanguagesAndOptionalFeatures\Micr
osoft-Windows-InternetExplorer-Optional-
Package~31bf3856ad364e35~amd64~~.cab"

Add-WindowsPackage -Online -PackagePath $packagePath

Restart-Computer
```

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Lab: Post installation configuration

1. Configure password, TCP/IP, domain join and time zone on Windows Server with Desktop Experience
2. Configure password, TCP/IP, domain join and time zone on Windows Server
3. Install app compatibility feature on demand

Required time: 30 minutes



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More information

Perform post-installation configuration of Windows Server - Learn | Microsoft Docs

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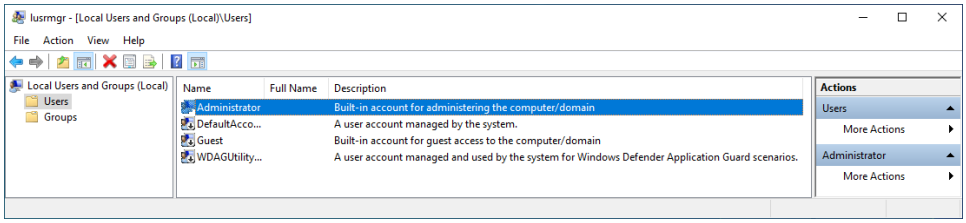
Managing users, computers and groups

Local users, Active Directory basics, domain users, groups, computers



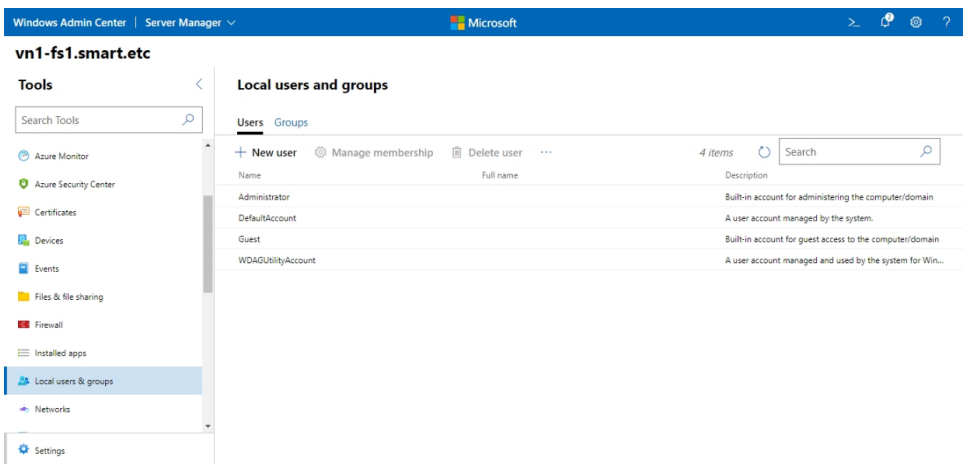
164

Managing local users using MMC



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Managing local users using Windows Admin Center



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Managing local users using PowerShell

List local users
`Get-LocalUser`
Create local user
`New-LocalUser`
Update local user (including password)
`Set-LocalUser`

Rename local user
`Rename-LocalUser`
Enable local user
`Enable-LocalUser`
Disable local user
`Disable-LocalUser`
Delete local user
`Remove-LocalUser`

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Practice: Manage local users

On CL1, CL2, and VN1-SRV5, create a local user with the name LocalAdmin. Use a different tool on each machine.
Required time: 5 minutes



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Active Directory Domain Services

A directory service that provides the following services in a Windows Server network:

- User account management
- User authentication
- Computer account management
- Access to networked resources
- Domain-wide services

Active Directory Domain Services Overview | Microsoft Docs

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Domain

Boundary of Security

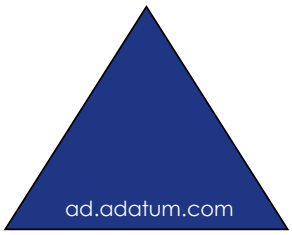
- Authentication
- Security Policies

Boundary of Replication

- Domain NC Replication

Boundary of DNS Namespace

Boundary of Administration

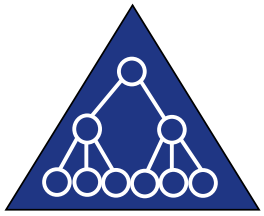


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Organizational Units

Containers within Domains
Distinct units of Administration
Unique to Domains

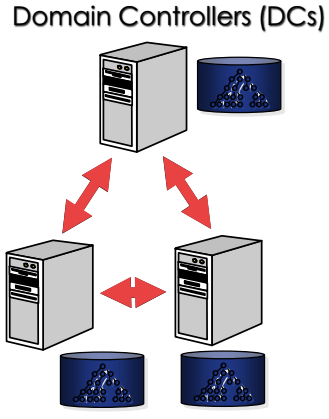


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Domain Controller Infrastructure

A Domain Controller hosts the Active Directory Database
Multiple Master Replication
Redundancy
Fault Tolerance



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Active Directory Objects

User
Computer
Group
Organizational Unit
Active Directory Accounts | Microsoft Docs

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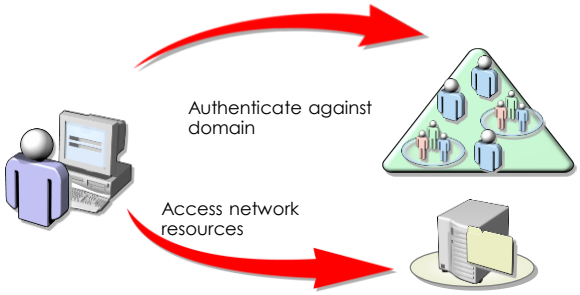
Active Directory Management Tools

You typically will perform AD DS management by using the following tools:

- Active Directory Administrative Center
- Active Directory Users and Computers
- Active Directory Sites and Services
- Active Directory Domains and Trusts
- Active Directory Schema snap-in
- Active Directory module for Windows PowerShell
- Windows Administrative Center

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How Does AD DS Work



1. User and computer objects are created in the directory
2. Groups of these objects then can be created
3. A client can use the user account to authenticate against AD DS
4. The user can try to access networked resources
5. The resources will again validate the authenticated user against AD DS

© ETC Jänner 25

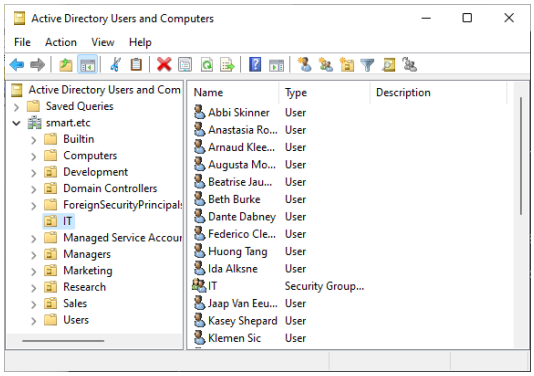
175

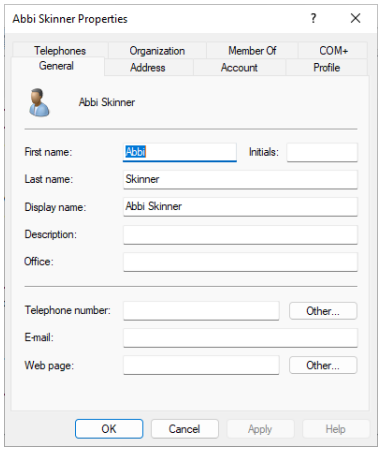
Identifying a user account

Attribute	Format	Uniqueness
GUID	80cede03-23e2-4d05-bbe4-4370545d59eb	Theoretically worldwide, practically forestwide
SID	S-1-5-21-4090364313-1945391572-4041509429-500	Domain-wide
SamAccountName	Smart\administrator	Domain-wide
UserPrincipalname	administrator@adatum.com	Forest-wide
Distinguished Name	cn=Administrator, cn=users, dc=ad, dc=adatum, dc=com	Forest-wide
Relative Distinguished Name	cn=Administrator	The container the object lives in

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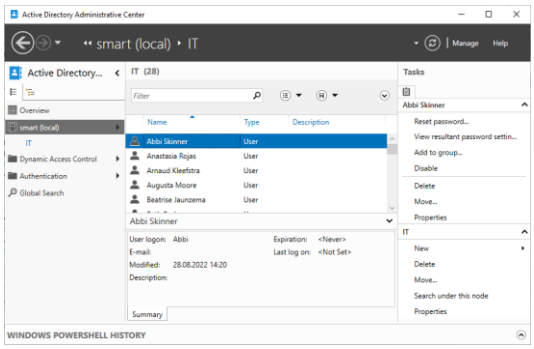
Managing domain users using Active Directory Users and Computers

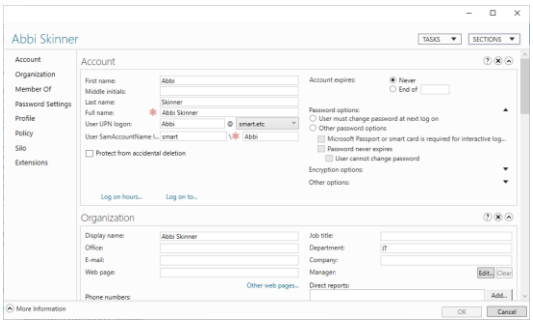




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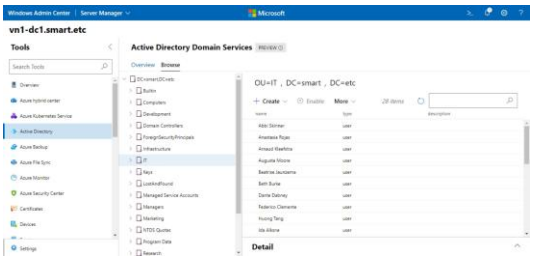
Managing domain users using Active Directory Administrative Center

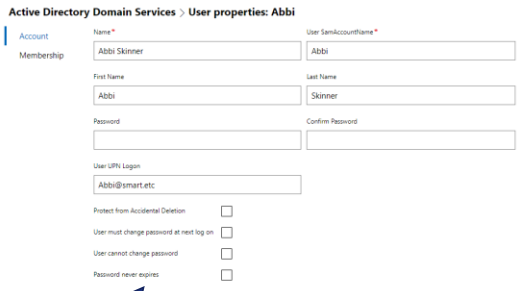




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Managing domain users using Windows Admin Center





Notice the missing Display name property. This is important for Microsoft Exchange. Therefore, Windows Admin Center will not be used for user management tasks.

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Managing domain users using PowerShell

Create a user
`New-ADUser`

Read users
`Get-ADUser -Filter $filter`

Update user
`Set-ADUser`

Rename user
`Rename-ADObject`

Reset password
`Set-ADAccountPassword`

Enable user
`Enable-ADObject`

Disable user
`Disable-ADObject`

Delete user
`Remove-ADUser`

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Active Directory PowerShell filter syntax

For `$filter` use the syntax

`{ propertyName operator value }`

Use PowerShell operators for *operator*

Put string values in quotes

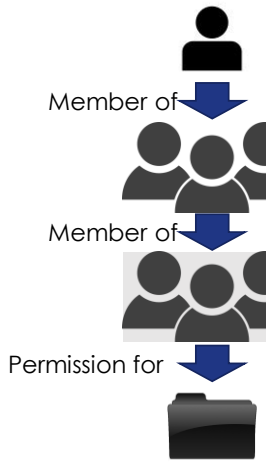
Examples

```
Get-ADUser -Filter `
    { Name -like 'Abbi*' -or Department -eq 'IT' }
$name = 'Abbi*'
$department = 'IT'
Get-ADUser -Filter `
    { Name -like $name -or Department -eq $department }
```

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Introduction to groups



Accounting
HR
Project Managers

Budgets Read
Budgets Modify

Often, entitling groups are created in an application, e.g. SQL roles or SharePoint groups

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Local groups

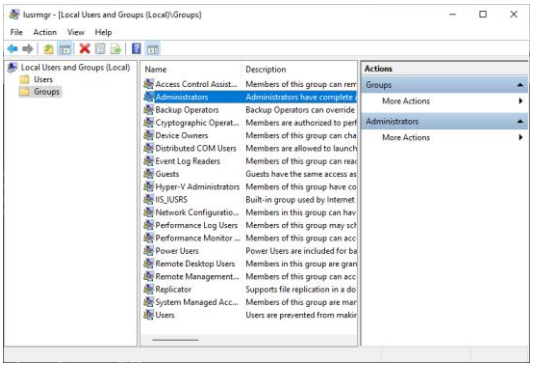
Exist on one computer only
Can contain identities and groups of all scopes from the computer‘s domain
Use for entitlement on one computer, e.g. Administrators group

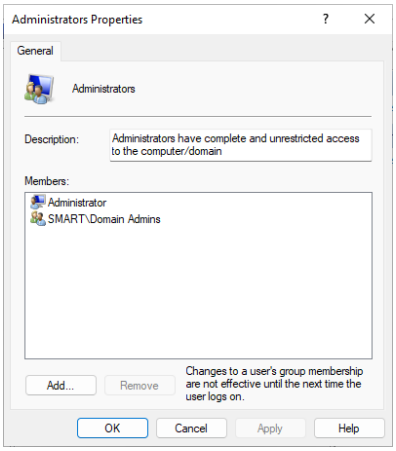


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Managing local groups using MMC

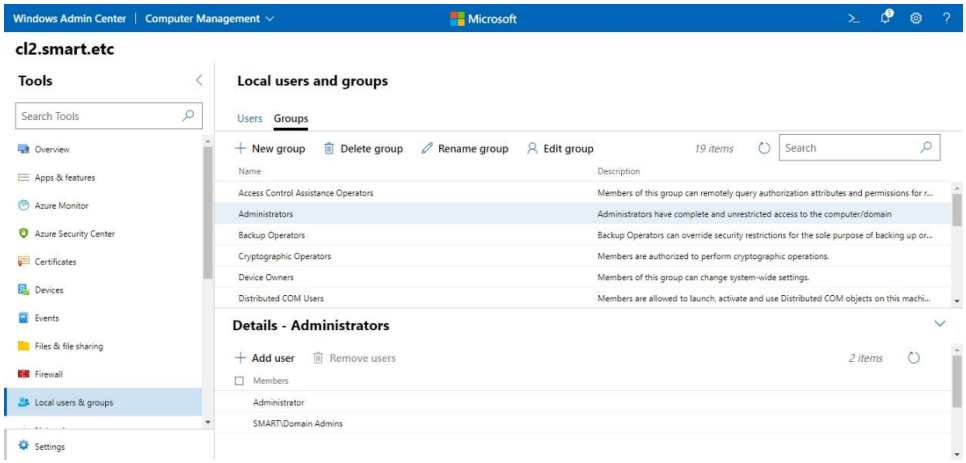




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Managing local groups using Windows Admin Center



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Managing local groups using PowerShell

Create local group
- `New-LocalGroup`

Read local groups
- `Get-LocalGroup`

Update local group
- `Set-LocalGroup`

Rename local groups
- `Rename-LocalGroup`

Delete local groups
- `Remove-LocalGroup`

Read local group members
- `Get-LocalGroupMembers`

Add local group members
- `Add-LocalGroupMembers`

Remove local group members
- `Remove-LocalGroupMembers`

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Practice: Manage local groups

On CL1, CL2, and VN1-SRV5, add LocalAdmin to the Administrators group. Use a different tool on each machine.
Required time: 5 minutes



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Domain group types

Distribution groups
- Used only with email applications
- Not security enabled (SID hidden)
- Cannot be given permissions



Security groups
- Security principal with a SID
- Can be given permissions
- Can also be email-enabled

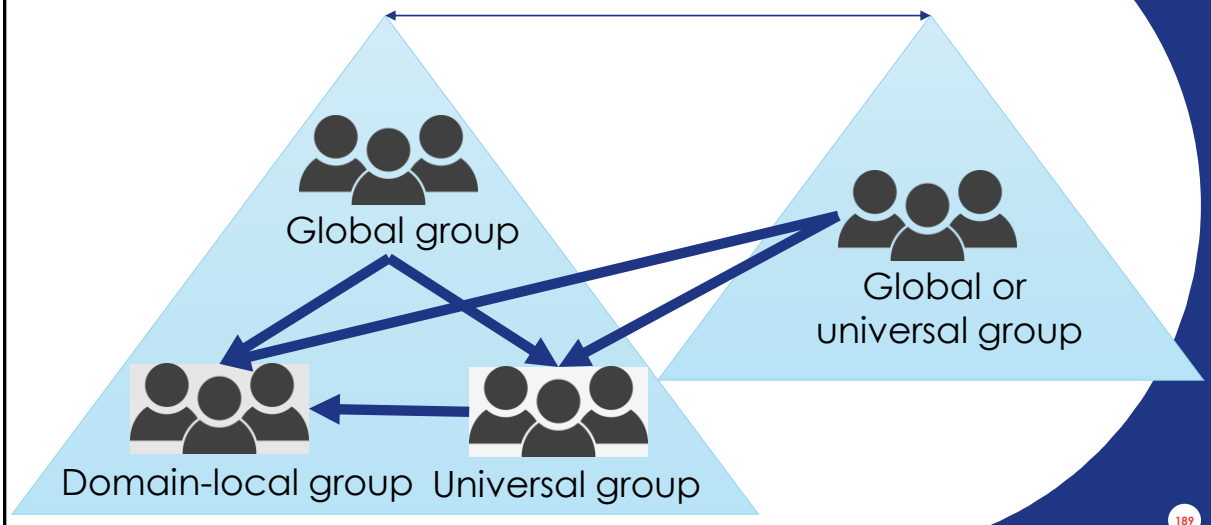


Note: Security groups and distribution groups can be converted to the other type of group

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Introduction to domain group scopes



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Domain group scope usage

In many environments, universal groups are to be avoided

Global groups for organizational groups
E.g., Accounting, HR, Project managers

Universal groups to group similar organizational groups in multi-domain environments
E.g., All Accounting, All HR, All Project managers

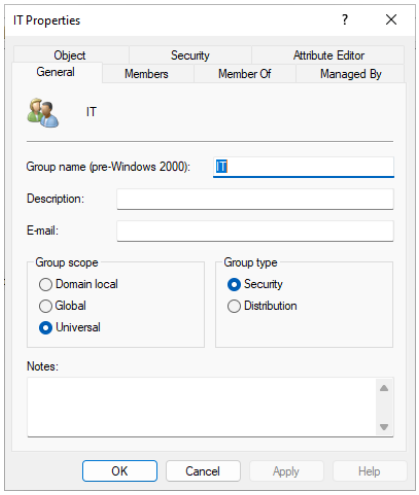
Domain-local groups for entitlement groups
E.g., Budgets read, Budgets modify

Domain-local groups are often replaced by application-specific groups
E.g., SQL Server roles, SharePoint groups

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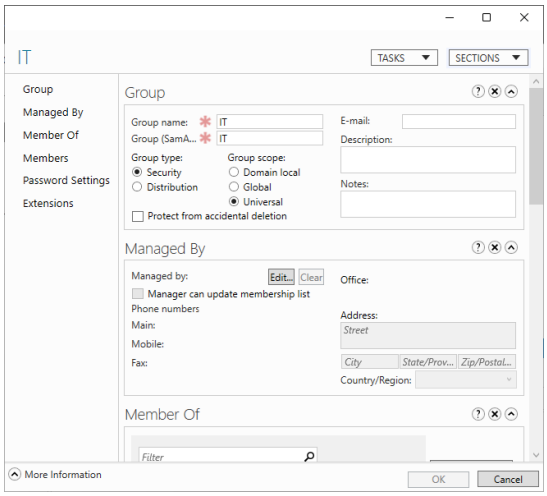
Managing domain groups using Active Directory Users and Computers



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Managing domain groups using Active Directory Administrative Center



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Managing domain groups using PowerShell

Create group
- New-ADGroup

Read groups
- Get-ADGroup

Update group
- Set-ADGroup

Rename group
- Rename-ADObject

Delete group
- Remove-ADGroup

Add group members
- Add-ADGroupMember

Read group members
- Get-ADGroupMember

Remove group members
- Remove-ADGroupMember

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Active Directory Recycle Bin

After enabled, deleted objects are retained for a number of days (tombstoneLifetime)

tombstoneLifetime is 60 days by default on domains created with Windows 2000 or Windows Server 2003

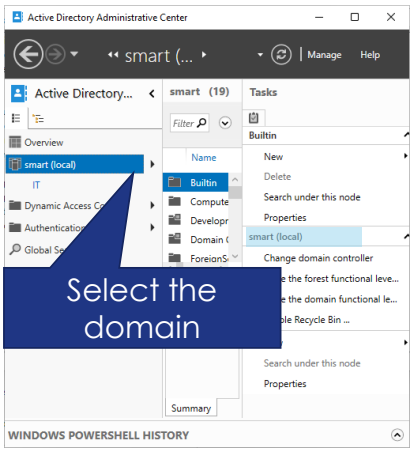
tombstoneLifetime is 180 days by default on all domains created with Windows Server 2003 SP1 or later

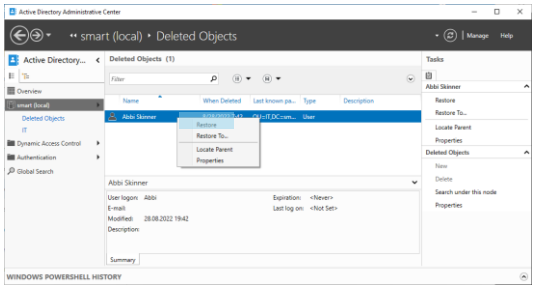
Advanced AD DS Management Using Active Directory Administrative Center (Level 200) | Microsoft Docs

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Enabling and using the Active Directory Recycle Bin





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Practice: Enable the Active Directory Recycle Bin

Enable the Active Directory using either Active Directory Administrative Center or PowerShell.

Required time: 5 minutes



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Computer accounts

A domain-joined computer trusts the domain

Domain-joined computers must authenticate with the domain

Therefore, every computer needs a computer account

- „Computer“ in the context of Active Directory, means Windows instance
- Dual-boot computers need two computer accounts!

Passwords for computer accounts are managed by the computer

Services running under Local System or Network Service accounts, use the computer account to authenticate to network resources

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Required permissions for domain join

Local Administrator rights

If no computer account exists: a new computer account is automatically created, if the user

- is a normal user in AD for a maximum of 10 computers (configurable)
- is a member of Domain Admin or Account Operators
- has permissions to create computer account objects in the default OU for new computers

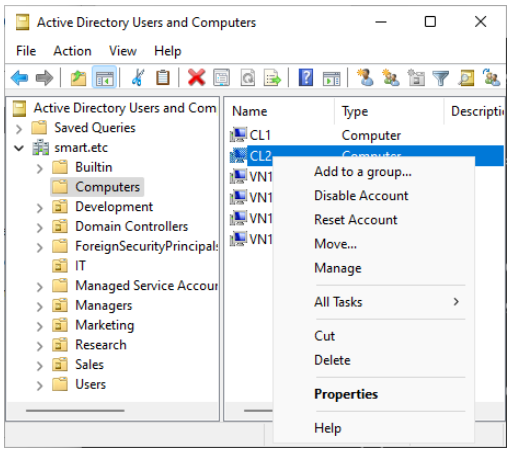
If computer account already exists, domain join requires the user to

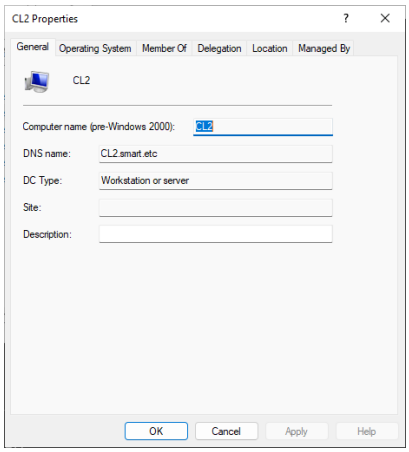
- be member of Domain Admin or Account Operators
- have the computer account created originally
- have special permission on the computer account or the OU containing it

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Managing computer accounts using Active Directory Users and Computers

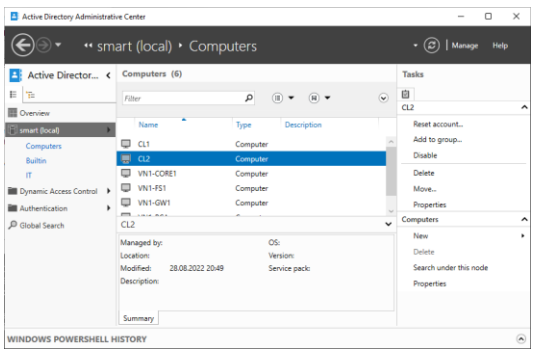


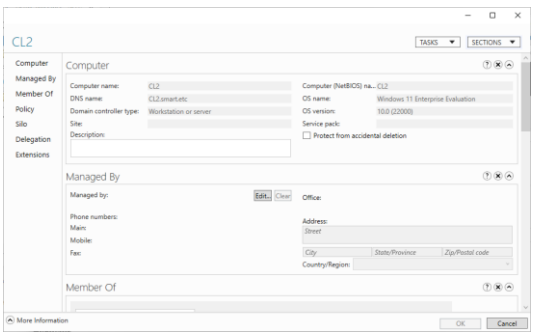


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Managing computer accounts using Active Directory Administrative Center





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Managing computer accounts using PowerShell

Create a computer
`New-ADComputer`
Read computers
`Get-ADComputer -Filter $filter`
Update computer
`Set-ADComputer`

Reset computer
`Set-ADAccountPassword`
Enable computer
`Enable-ADObject`
Disable computer
`Disable-ADObject`
Delete computer
`Remove-ADComputer`

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Troubleshooting the trust relationship of a computer

If the password on the computer and in Active Directory are out of sync, the trust relationship breaks, and domain users cannot login anymore.

Use one of the following commands to reset the computer account password and restore the trust relationship:

- ```
  Test-ComputerSecureChannel `
      -Repair
      -Credential $credential
  ```
- `Reset-ComputerMachinePassword -Credential $credential`

For `$credential`, use the name of a domain account with appropriate permissions.

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  ```

Lab: Manage domain users, groups, and computers

1. Manage domain users
2. Manage domain groups
3. Manage domain computers

Required time: 45 minutes



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More information

Active Directory Domain Services | Microsoft Docs

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Managing local storage

Disks, partitions, volumes, NTFS, ReFS, RAID, mount points, links



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Virtual disks

Virtual hard disks (vhd or vhdx) are files that you can use like physical hard disks

Can be created using

- Disk Management
- Windows Admin Center
- diskpart.exe
- Windows PowerShell
- Hyper-V Manager

VHD cmdlets are only available with Hyper-V installed. This will not be covered in this module.

Virtual and physical machines can boot from virtual disks

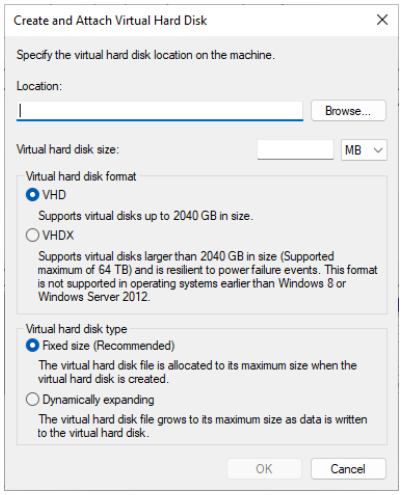
Can be used as deployment technology

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Creating virtual disks using Disk Management

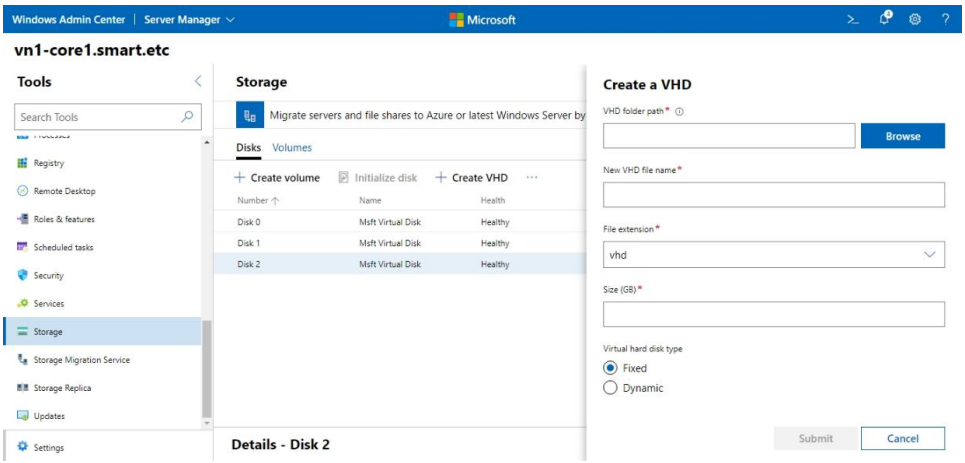
On the menu, click **Action, Create VHD**



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Create virtual disks using Windows Admin Center



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Selecting a partition table format

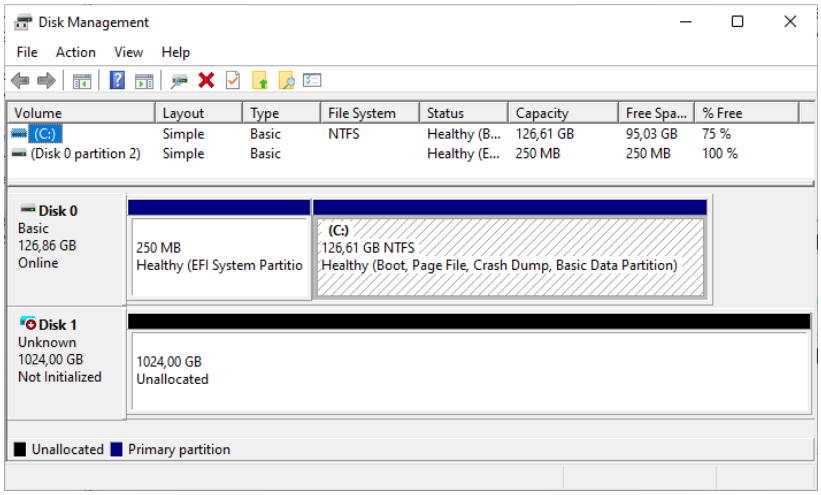
MBR
- Standard Partition table format since early 1980s
- Supports a maximum of 4 primary partitions per drive
- Can partition a disk up to 2 terabytes

GPT
- GPT is the successor of MBR partition table format
- Supports a maximum of 128 partitions per drive
- Can partition a disk up to 18 exabytes

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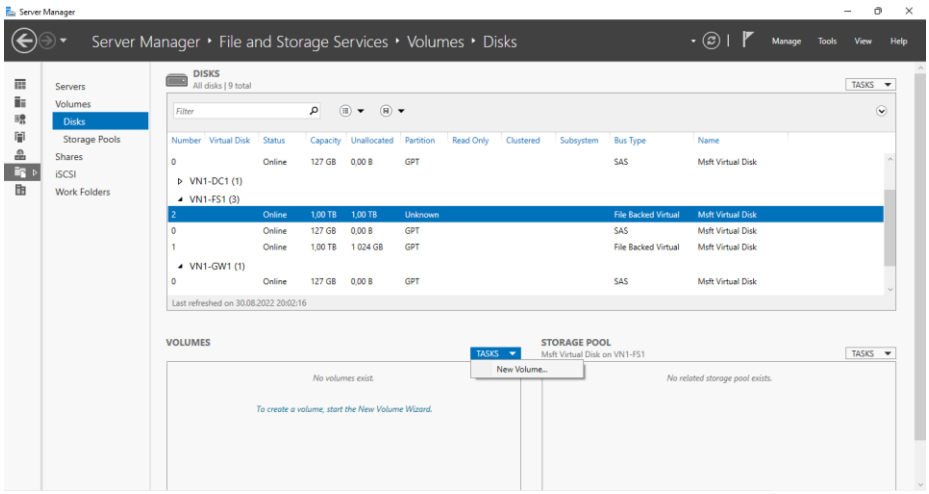
Managing disks and volumes using Disk Management



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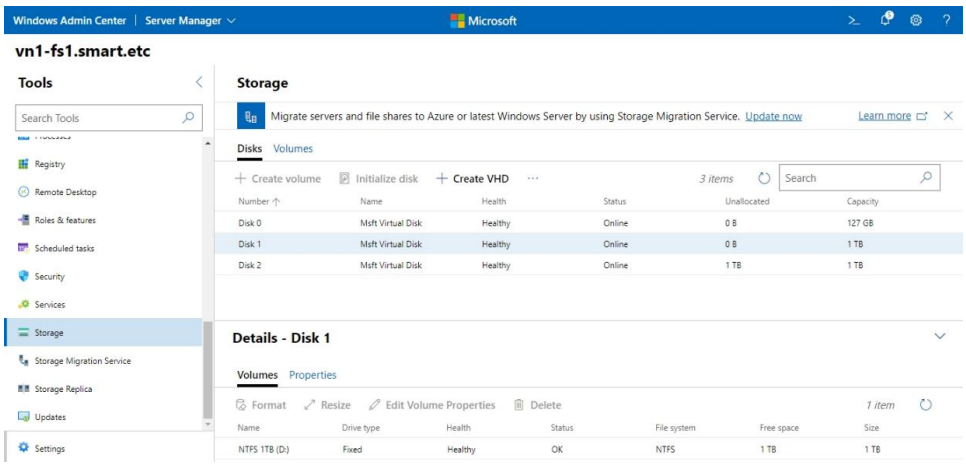
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Managing disks and volumes using Server Manager



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Managing disks and volumes using Windows Admin Center



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Managing disks and volumes using PowerShell

Initialize disk
- `Initialize-Disk`

Reset disk
- `Clear-Disk`

Create partition
- `New-Partition`

List partitions
- `Get-Partition`

Change partition, e.g., drive letter
- `Set-Partition`
- `Set-Volume`

Add drive letters, mount points, etc.
- `Add-PartitionAccessPath`

Change the size
- `Resize-Partition`

Delete partition
- `Remove-Partition`

Format the volume
- `Format-Volume`

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Partition vs. Volume

Partition

Logical division of a disk

Does not need to have a file system

Maximum size is the size of a single disk

Volume

Single, accessible storage area with a single file system

Consists of one or more partitions

Can span multiple disks

Maximum size is only limited by the file system

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What is Resilient File System (ReFS)?

The Resilient File System (ReFS) is Microsoft's newest file system, designed to maximize data availability, scale efficiently to large data sets across diverse workloads, and provide data integrity with resiliency to corruption. It seeks to address an expanding set of storage scenarios and establish a foundation for future innovations.

Resilient File System (ReFS) overview | Microsoft Docs

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ReFS benefits

Resiliency
- Integrity-streams: checksums for metadata and optionally for file data
- Storage Spaces integration: automatic repairs on mirror or parity spaces
- Salvaging data: automatic removal of corrupt data from namespace
- Proactive error correction: periodic scrubbing scans volume, identifies corruptions and triggers repair

Performance
- Mirror-accelerated parity
- Accelerated VM operations
  - Block cloning: quick merge operations
  - Sparse VDL: creation of fixed VHDs take seconds instead of 10s of minutes

Scalability

Supports extremely large data sets

Resilient File System (ReFS) overview | Microsoft Docs

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ReFS supported deployments

Basic disks and backup target

- Best suited for applications that implement their own software resiliency and availability solutions
- Applications that introduce their own resiliency and availability software solutions can use integrity-streams, block-cloning

Storage Spaces

Storage Spaces Direct

Not supported as boot and system volume

Resilient File System (ReFS) overview | Microsoft Docs

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ReFS and NTFS limits

Feature	ReFS	NTFS
Maximum file name length	255 Unicode characters	255 Unicode characters
Maximum path name length	32K Unicode characters	32K Unicode characters
Maximum file size	35 PB (petabytes)	256 TB
Maximum volume size	35 PB	256 TB

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ReFS unsupported features

File system compression
File system encryption
Transactions
Object Ids
Offloaded Data Transfer (ODX)
Short names
Extended attributes
Disk quotas
Bootable
Page file support
Supported on removable media

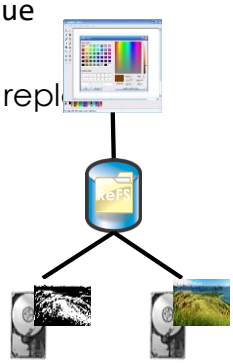
221

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ReFS Auto Correction

```
Set-FileIntegrity FileName $filename -Enable $true
```

1. Application accesses data
2. ReFS volume detects corruption and automatically repl
   the corrupt data with correct data
3. Mirror space offers all data copies to the file system



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What Is RAID?

RAID combines multiple disks into a single logical unit to provide fault tolerance and performance

Provides fault tolerance by using:

- Disk mirroring
- Parity information

Can provide performance benefits by spreading disk I/O across multiple disks

Can be configured using several different levels

Should not replace server backups

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Extending and Shrinking Volumes

You can resize NTFS volumes from the operating system, beginning with Windows Vista and Windows Server 2003

When you want to resize a disk, consider the following:

- You can extend or shrink NTFS volumes
- ReFS volumes can only be extended
- FAT/FAT32/exFAT cannot be resized
- To extend, the free space must be adjacent
- You can shrink a volume only up to immovable files
- Bad clusters on a disk will prevent you from shrinking a volume

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What Are Mount Points?

A mount point is a reference to a location on a disk that enables operating system access to disk resources

Volume mount points enable you to mount volumes or disks as folders rather than using drive letters

Use volume mount points when:

- You need to add disk space without changing the folder structure
- You do not have drive letters available for creating new volumes

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What are links?

A link file is special type of file that contains a reference to another file or directory

Link options:

- Hard link
- Junction
- Symbolic Link (soft link)

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Hard link

Processed by the file system
Link to files on the same volume only
Like an additional folder entry for the same file
Deletion of the original file entry, does not delete the file; the hard link still works

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Junction

Processed by the file system
Link to folders on any volume of the same system
Similar to mount points
Use for deep links into the file system on one computer

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Symbolic link (soft link)

Processed by the client
Link to folders on any volume of the same system (but network access will fail)
Link to shares on any system

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Link type comparison

	Hard link	Junction	Symbolic link
Target	Files on same volume	Folder on same computer	Folder on same computer Share on other computer
Example target	Link -> C:\folderB\file2.txt	Link -> D:\folderD	1. Link -> D:\folderD 2. Link-> \\server\folderD
Local access result	C:\folderB\file2.txt	D:\folderD	1. D:\folderD 2. \\Server\folderD
Network access to \\share\link result	C:\folderB\file2.txt	D:\folderD	1. Error 2. \\Server\folderD

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Managing links with PowerShell

```
New-Item `
    -ItemType HardLink` 
    -Target $targetFile `
    -Name $name `
    -Path $path `
New-Item `
    -ItemType Junction`
    -Target $targetFolder `
    -Name $name `
    -Path $path `
```

```
New-Item `
    -ItemType SymbolicLink`
    -Target $targetShare `
    -Name $name `
    -Path $path `
```

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Discussion

One of the volumes on a server a filling up. To provide additional disk space, you add an additional disk to the server.

How would you provide the additional disk space to the users of the server?

- Create a new volume and move files to the new volume?
- Extend the existing volume?
- Create a new volume and create a mount point on the existing volume?
- Are there other options?

Required time: 10 minutes



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Lab: Manage local storage

1. Manage disks
2. Manage volumes
3. Manage mount points
4. Manage links and junctions

Required time: 40 minutes



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More information

Windows Server Storage documentation | Microsoft Docs

234

Introduction to Public Key Infrastructure

Applications, algorithms, certificate management



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Public Key Infrastructure (PKI)

- Is a standard approach to security-based tools, technologies, processes, and services used to enhance the security of communications, applications, and business transactions
- Relies on the exchange of digital certificates between authenticated users and trusted resources
- PKI provides:
  - Confidentiality
  - Integrity
  - Authenticity
  - Non-repudiation

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PKI applications

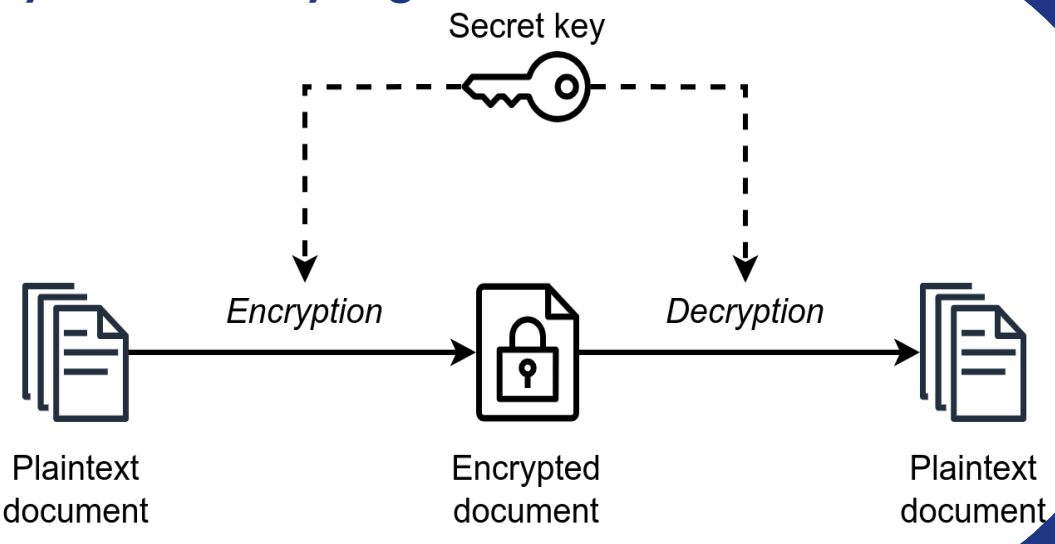
PKI is used when encryption or digital signatures are required, e.g.,

- Secure network traffic (e.g., TLS, VPN)
- Encrypting File System
- Bitlocker
- Client authentication
- Rights management

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Symmetric key algorithm

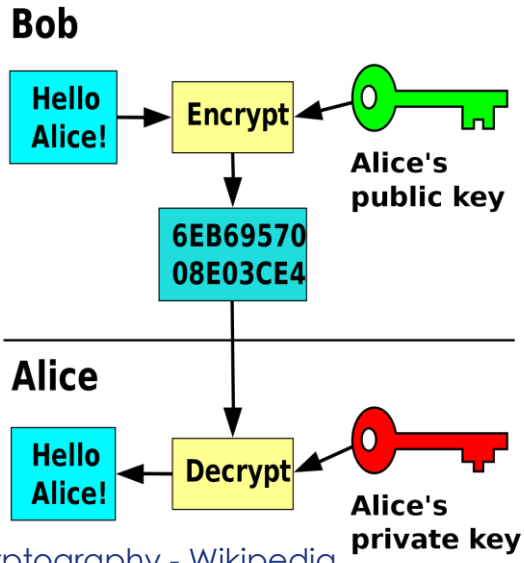


Symmetric-key algorithm - Wikipedia

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Public-key cryptography

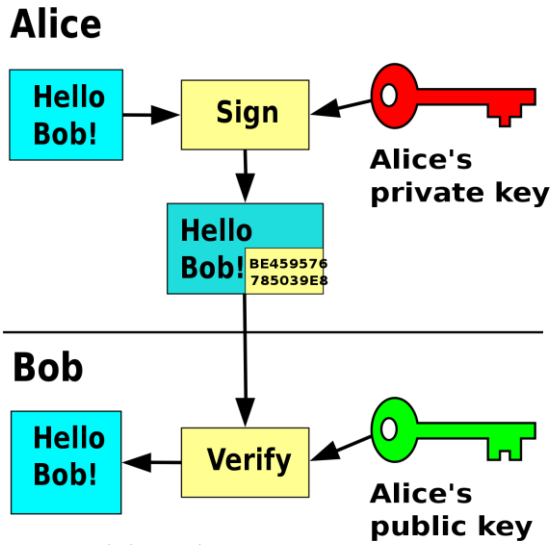


Public-key cryptography - Wikipedia

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Digital signature

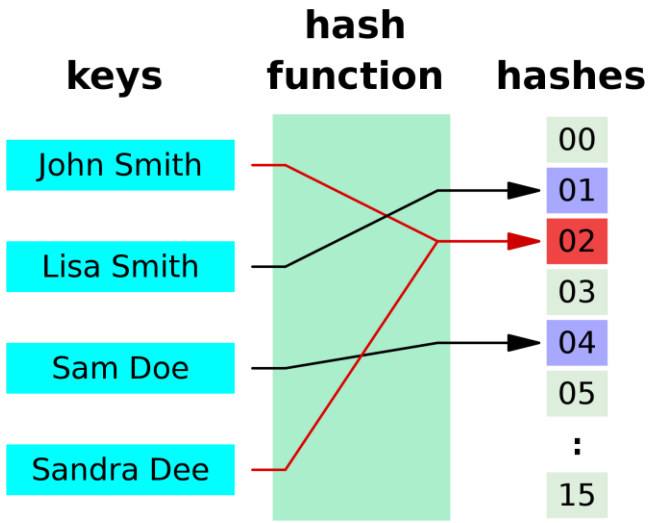


Digital signature - Wikipedia

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Hash function



Hash function - Wikipedia

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Symmetric keys, asymmetric keys and hash function in the context of PKI

Symmetric keys
- Used to encrypt data
- Can use large keys (>= 4096 Bits) without affecting performance too much
- Robust against brute-force guessing attacks, even if parts of the content are known

Hash-functions
- Used as „check sums“ for data

Asymmetric keys
- Used to encrypt symmetric keys or hash values
- Use shorter keys (typically 256 Bits)
- Vulnerable against brute-force guessing attacks, if parts of the content are known

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Certificates

- Associated with a private key
  - Private key is stored separately on the computer/in the user profile only
- Contains public key
- Is digitally signed by certification authority or self-signed

- Other important fields:
  - Subject name
    CN of the subject, e.g. FQDN
  - Subject Alternative Name (SAN)
    - Alternative FQDNs
    - Requirement for modern web browsers, even if subject name equals FQDN
  - Purpose, e.g.
    - Client authentication
    - Server authentication
    - Code signing
  - Valid from/to
  - Thumbprint

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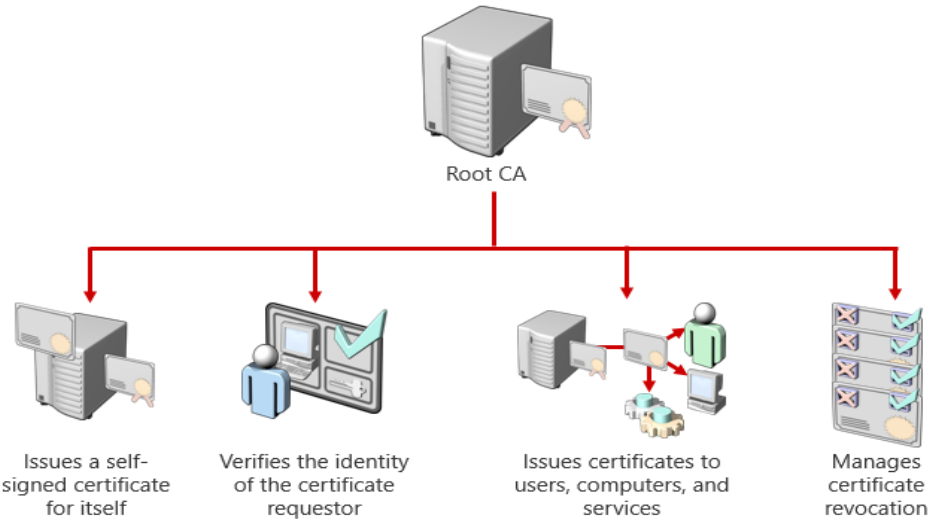
Components of a PKI Solution

- Certificate Authority (CA)
  - Issues signed certificate based on certificate templates added to the CA
  - Publishes a certificate revocation list
- Digital Certificates
  - Are issued to clients (user/computer)
  - Based on certificate requests generated by the client

- Certificates Templates
  - Define parameters, like purpose, encryption algorithm, validity period, for issued certificates
  - Stored in Active Directory
- Certificate revocation list (CRL)
- CRL Distribution point (CDP)
- PKI-enabled applications
- Management Tools
- Authority Information Access (AIA)
  - Extension in certificates
  - Location for information about the issuer (CA) of the certificate

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Certification Authority



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Public vs. private CAs

Public CA
- Trusted by default by common operating systems and web browsers
- Free and paid offers

Private CA
- Must be added manually to Trusted Root Certification Authorities
- Domain members trust Enterprise CAs automatically

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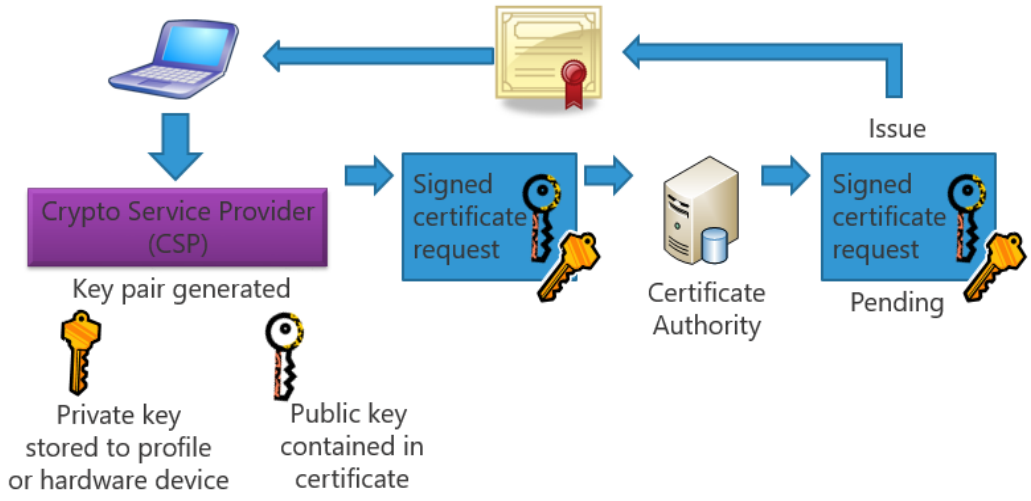
Standalone vs. Enterprise CA

Standalone CAs	Enterprise CAs
Must be used if any CA (root/intermediate/policy) is offline because a standalone CA is not joined to an AD DS domain	Requires the use of AD DS and stores information in AD DS
	Can use Group Policy to propagate certificates to the trusted root CA certificate store
Users must provide identifying information and specify the type of certificate	Publishes user certificates and CRLs to AD DS
Does not support certificate templates	Issues certificates based on a certificate template
All certificate requests are kept pending until administrator approval	Supports autoenrollment for issuing certificates

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Certificate Enrollment



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Requirements for a successful certificate request with an Enterprise CA

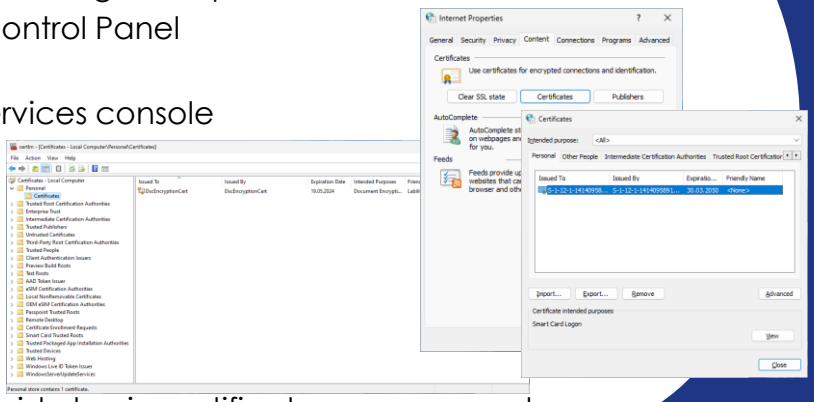
- The certificate template must have been made available on the CA
- User certificate
  - User must have read permission on certificate template
  - User must have enroll permission on certificate template
- Computer certificate
  - User must have local administrator rights
  - Computer must have read permission on certificate template
  - Computer must have enroll permission on certificate template

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Certificate management tools

- Certificates snap-in for MMC
  Tip: run `certlm.msc` to manage computer certificates
- Internet Properties in Control Panel
  User certificates only
- Internet Information Services console
  IIS certificates only
- `certutil.exe`
- PowerShell
  - `Get-Certificate`
  - `Export-Certificate`
  - `Import-Certificate`
- Many applications provide basic certificate management capabilities



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Requirements for a valid certificate

- Issuer (CA) must be trusted
- Subject name or subject alternative name must match the subject (requested service/FQDN)
  For modern web browsers, the SAN must match the FQDN
- Current date and time must be greater than valid from and less than valid to
- Certificate's purpose must match the intended purpose
- CRL must be accessible
- Certificate must not be on the CRL

Baseline Requirements Documents (SSL/TLS Server Certificates) – CAB Forum

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More information

certutil | Microsoft Learn

Course smart Public Key Infrastructure (PKI) unter Windows Server - Aufbau und Praxis

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Managing permissions and shares

Permissions, shares, offline files, volume shadow copies



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How do permissions work?

Permissions define the type of access granted to a user or group for an object or object property

Permissions can be granted to:

- Groups, users, and other objects with security identifiers in the domain
- Groups and users in a specific domain and any trusted domains
- Local groups and users on the computer where the object resides

Permissions Types

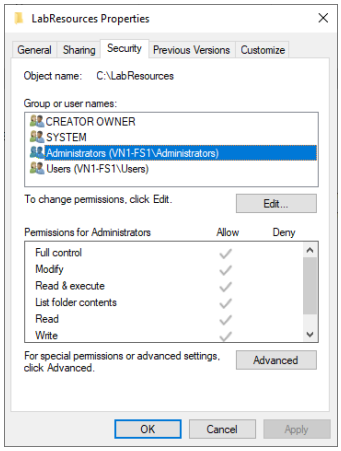
- Read
- Modify
- Change owner
- Delete

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What are NTFS (ReFS) permissions?

- Are assigned per file or directory
- Can always be set by the file or folder owner
- Are much more flexible than share permissions
- Can be allowed or denied
- Permissions are accumulated
- Deny takes precedence over Allow (with exceptions)



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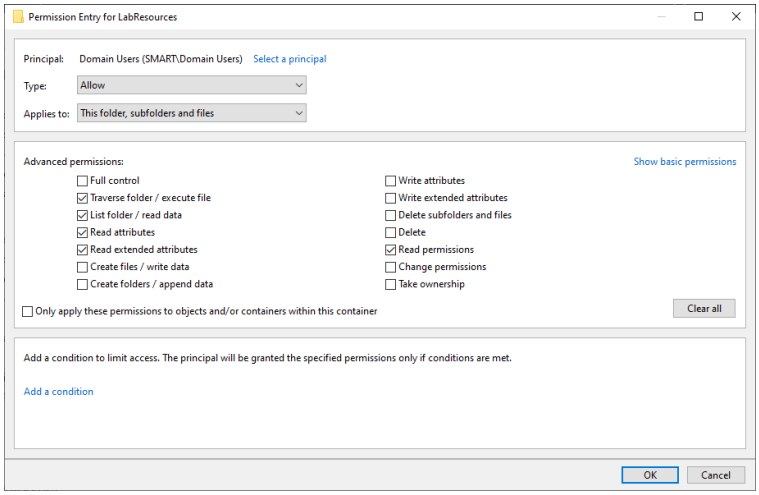
Basic permissions

Special Permissions	Full Control	Modify	Read & Execute	List Folder Contents	Read	Write
Traverse Folder/Execute File	x	x	x	x		
List Folder/Read Data	x	x	x	x	x	
Read Attributes	x	x	x	x	x	
Read Extended Attributes	x	x	x	x	x	
Create Files/Write Data	x	x				x
Create Folders/Append Data	x	x				x
Write Attributes	x	x				x
Write Extended Attributes	x	x				x
Delete Subfolders and Files	x					
Delete	x	x				
Read Permissions	x	x	x	x	x	x
Change Permissions	x					
Take Ownership	x					
Synchronize	x	x	x	x	x	x

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Advanced permissions



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Explicit and Inherited Permissions

Explicit Permissions
- Permissions set by default on non-child objects while object creation, or by user action on non-child, parent, or child objects

Inherited Permissions
- Permissions propagated to an object from a parent object and ease the task of managing permissions and ensure consistency of permissions among all objects within a given container

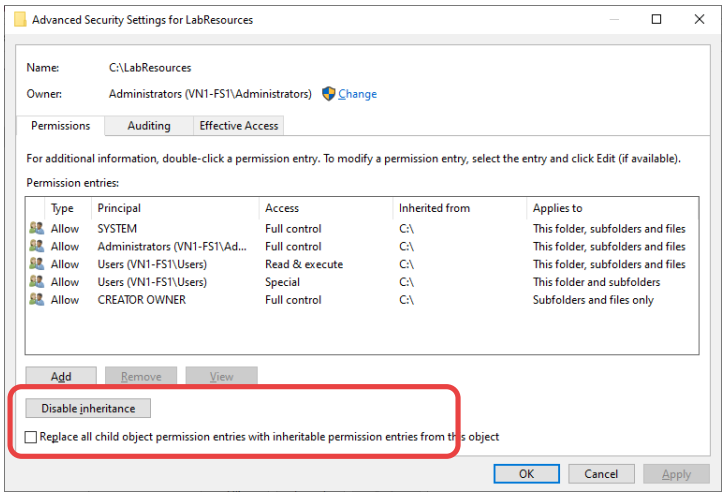
*Inherited **Deny** permissions do not prevent access to an object if the object has an Explicit **Allow** permission entry. Explicit permissions take precedence over Inherited permissions, even Inherited **Deny** permissions*

Three recommended ways to make changes to Inherited Permissions
- Make the changes to the parent object where the permissions are explicitly defined, and the child object inherits these permissions
- Select the **Allow** Permission to override the Inherited **Deny** permission
- Clear the Include Inheritable permissions from this object's parent check box and make changes to the permissions or remove users or groups from the permissions list

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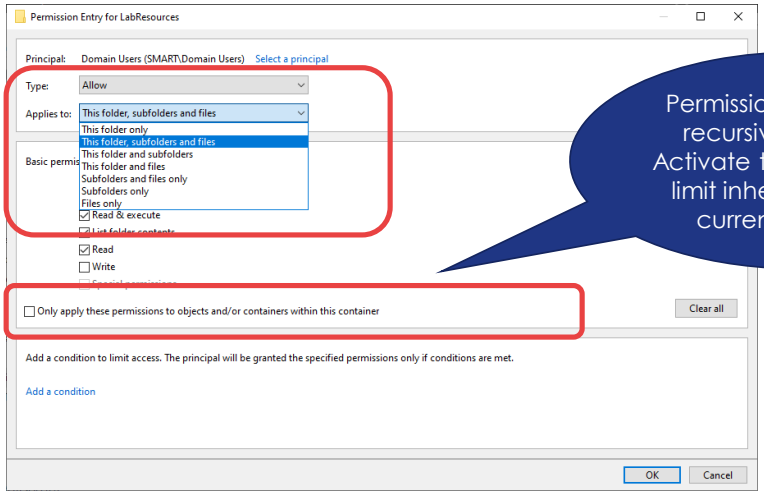
259

Enable/Disable inheritance



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Manage inheritance



Permissions are applied recursive by default. Activate this checkbox to limit inheritance to the current container.

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Manage permissions using PowerShell

1. Get the ACL for an object
   `$acl = Get-ACL -Path $path`
2. Modify the ACL

• Manage inheritance
• Manage access rules

3. Write back the ACL
   `$acl | Set-Acl –Path $path`

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Manage inheritance using PowerShell

Disable inheritance, preserve existing access rules
`$acl.SetAccessRuleProtection($true, $true)`
Disable inheritance, remove all inherited access rules
`$acl.SetAccessRuleProtection($true, $false)`
Enable inheritance
`$acl.SetAccessRuleProtection($false, $false)`
or
`$acl.SetAccessRuleProtection($false, $true)`
Note: The second parameter is ignored.

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Manage access rules using PowerShell

Read access rules

```
$acl.Access
```

Add access rule

```
$acl.AddAccessRule($accessRule)
```

Update an access rule

```
$acl.SetAccessRule($accessRule)
```

Remove an access rule

```
$acl.RemoveAccessRule($accessRule)
```

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Create access rules using PowerShell

```
$accessRule = New-Object `
    -TypeName `
        System.Security.AccessControl.FileSystemAccessRule `
    -ArgumentList `
        $identity, `
        $fileSystemRights, `
        $inheritanceFlags, `
        $propagationFlags, `
        $type
```

`$identity` is an identifier of a group or user account

```
E.g., 'smart\Domain Users'
```

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Access rule parameters

FileSystemRights
FileSystemRights Enum (System.Security.AccessControl) | Microsoft Docs
InheritanceFlags
InheritanceFlags Enum (System.Security.AccessControl) | Microsoft Docs
PropagationFlags
PropagationFlags Enum (System.Security.AccessControl) | Microsoft Docs
Type
AccessControlType Enum (System.Security.AccessControl) | Microsoft Docs
You can use strings. Multiple strings can be combined using a comma-separated list. E.g.

- `$fileSystemRights = 'Read, Write'`
- `$inheritanceFlags = 'ContainerInherit, ObjectInherit'`

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Managing permissions using PowerShell example

```
$identity = 'smart\Training Modify'
$fileSystemRights = 'Modify'

# Apply to folders and files
$inheritanceFlags = `
    'ContainerInherit, ObjectInherit'

# Recursive application
$propagationFlags = 'None'

$accessControlType = 'Allow'
```

```
$acl = Get-Acl -Path $path
$acl.SetAccessRuleProtection($true, $true)
$acl.Access |
Where-Object {
    $PSItem.IdentityReference `
    -eq 'BUILTIN\Users'
} |
ForEach-Object {
    $acl.RemoveAccessRule($PSItem)
}
$accessRule = New-Object `
    -TypeName `
        System.Security.AccessControl.FileSystemAccessRule `
    -ArgumentList `
        $identity, `
        $fileSystemRights, `
        $inheritanceFlags, `
        $propagationFlags, `
        $accessControlType
$acl.AddAccessRule($accessRule)
$acl | Set-Acl -Path $trainingPath
```

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Impact of copying and moving files and folders on set permissions

When you copy or move a file or folder to a different NTFS volume…

The file or folder inherits permissions from the destination folder

When you copy or move a file or folder within an NTFS volume…

The file or folder inherits permissions from the new parent folder

When moving a file or folder that has explicitly assigned permissions, those permissions are retained in addition to the newly inherited permissions

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Managing object ownership

Object ownership

Every object has an owner that controls how permissions are set on it and to whom permissions are granted

Ownership can be taken by:

- An administrator group given the Take ownership of files or other objects user right
- Any user or group having the Take Ownership permission on the object
- A user having the Restore files and directories user right

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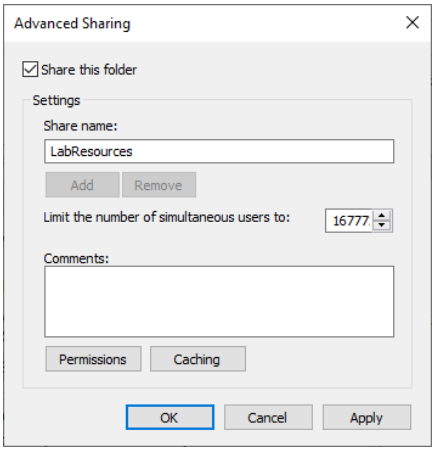
What are shares?

Folders configured to be accessible on the network

Share permissions controlling the access over the network

Additional share options

- Encryption of SMB traffic
- Access-based enumeration
- Offline access



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Hidden shares and default shares

Hidden shares

Append a $ sign to the share name to hide the share from File Explorer

Users knowing the share name can still access it

No impact on permissions

Default shares

Volumes on non-removable storage

C$, D$, E$, …

Windows folder

ADMIN$

Additional default shares on Domain Controllers only

- NETLOGON
- SYSVOL

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What are share permissions?

Control access to folders shared over the network
Do not control local access to files and folders
Can be allowed or denied

Permission	Description
Full Control	Allows all permissions including the ability to change permissions
Read	Allows users to read existing files
Change	Allows users to create new files or delete, modify, and read existing files

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Effective permissions

When combining shared folder and NTFS permissions, the most restrictive permission is applied

Example:

If a user or group is given the shared folder permission of Read and the NTFS permission of Write, the user or group will only be able to read the file because it is the more restrictive permission.

To get access to a resource, both the share and the NTFS file and folder permissions must have set the correct permissions

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Discussion

How would you set share permissions?

- Everyone Full Control
- As restrictive as possible

Required time: 5 minutes



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Permission scenarios

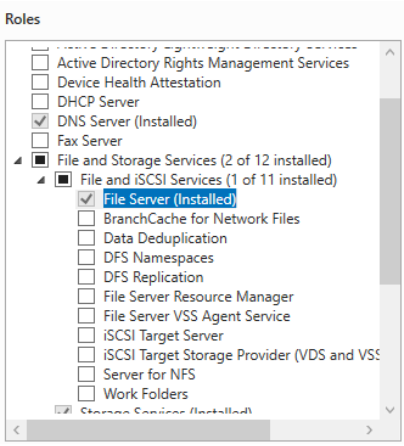
Folder Type	Share Permissions	NTFS Permissions
Public Folder - A folder that can be accessed by everyone	Grant **Change** permission to the Users group	Grant **Modify** permission to the Users group
Drop Folder - A folder where users can drop confidential reports or homework assignments that only the group manager or instructor can read	▪ Grant **Change** permission to the Users group ▪ Grant **Full Control** permission to the group manager	▪ Grant **Write** permission for the Users group that is applied to **This Folder** only. ▪ Grant Full Control permission to the group manager
Application Folder - A folder containing applications that can be run over the network	Grant **Read** permission to the Users group	Grant **Read**, **Read & Execute**, **and List Folder Contents** permissions to the Users group
Home Folder - An individual folder for each user. Only the user has access to the folder	Grant **Full Control** permission to each user on his or her respective folder.	Grant **Full Control** permission to each user on his or her respective folder.

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Prerequisites for shares

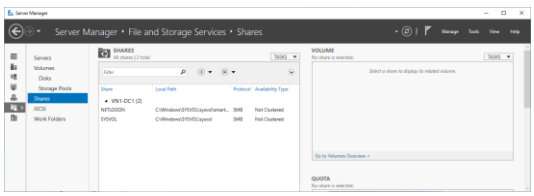
Install the File Server role
or
Configure the Windows Defender Firewall

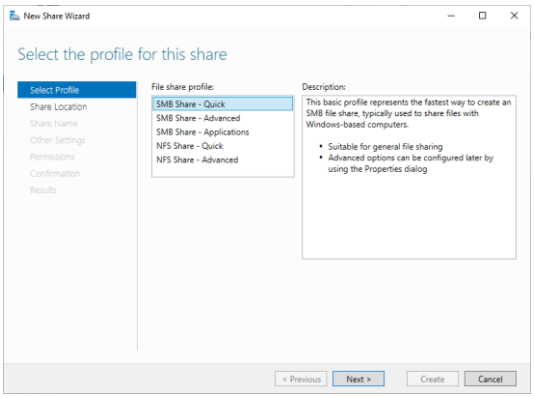


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Manage shares using Server Manager

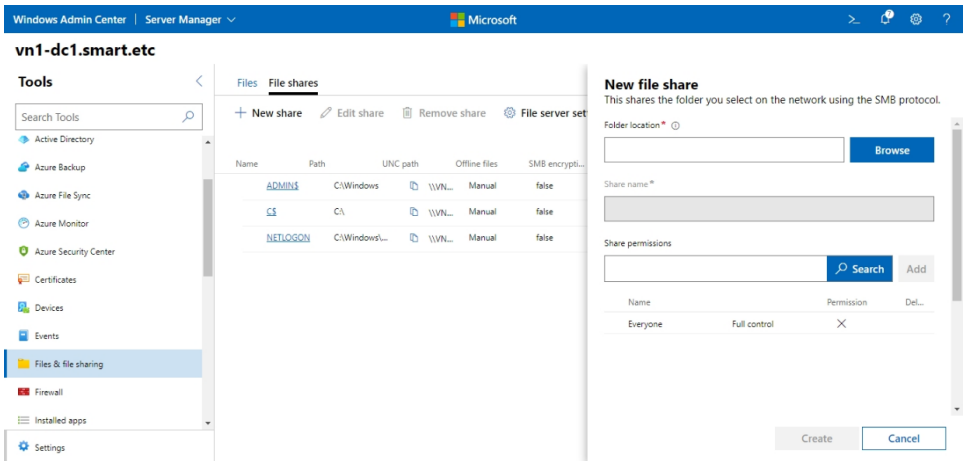




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Manage shares using Windows Admin Center



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Manage shares using Powershell

Create share

`New-SmbShare`

Read shares

`Get-SmbShare`

Update share, e.g., share permissions

`Set-SmbShare`

Delete share

`Remove-SmbShare`

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What is access-based enumeration?

Controls the visibility of shared folders according to the permissions set on the shared folder

Available for shared folders only

Configurable on a per shared folder basis

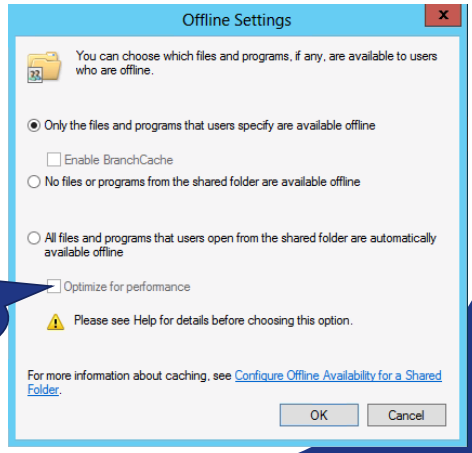
280

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What are offline files?

Offline file settings allow client computer to cache network files locally for offline use when they are disconnected from the network

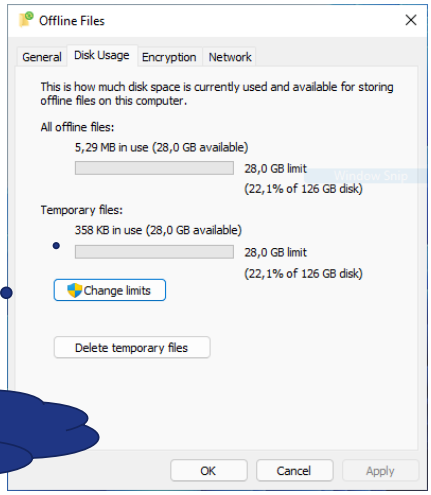
Optimize for performance has no effects on modern Windows versions (starting from Vista)



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Offline file settings on clients



Temporary files are automatically cached

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What Are Shadow Copies?

Allow access to previous versions of files

Are based on tracking disk changes

- Disk space is allocated on the same volume
- When the space is full, older shadow copies are removed

Are not a replacement for backups

Are not suitable for recovering databases



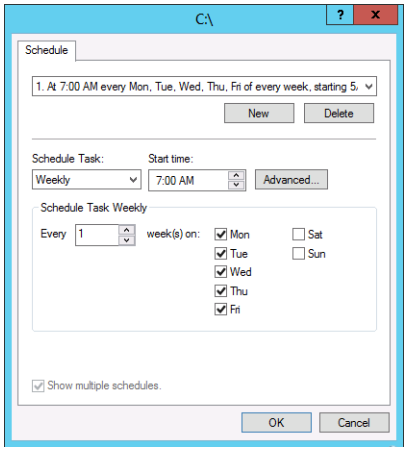
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Considerations for Scheduling Shadow Copies

- Default schedule is 7:00 A.M. and noon
- Create a shadow copy schedule based on:
  - Volume of changes
  - Importance of changes
  - Storage limitations





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Restoring Data from a Shadow Copy

- Previous versions are accessible from the **Properties** window of a file or folder
  - Administrators can restore previous versions directly on the server
  - Users can restore previous versions over the network
- All users can:
  - Restore a file or folder
  - Copy a file or folder to an alternate location
  - Browse previous versions to select the correct version



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Lab: Manage file sharing

1. Manage file shares and permissions
2. Explore offline files
3. Explore Volume Shadow Copies

Required time: 60 minutes



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More information

Access Control: Understanding Windows File And Registry Permissions | Microsoft Docs

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Introduction to Group Policies

Overview, Group Policies Management Console, Group Policies Objects, Group Policies Links



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What is configuration management?

Configuration management is a centralized approach to applying one or more changes to more than one user or computer

The key elements of configuration management are:

- Setting
- Scope
- Application

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Benefits of using Group Policies

Group Policies (actually: Group of Policies) is a very powerful administrative tool for configuration management

You can use it to enforce various types of settings to a large number of users and computers

Typically, you use GPOs to:

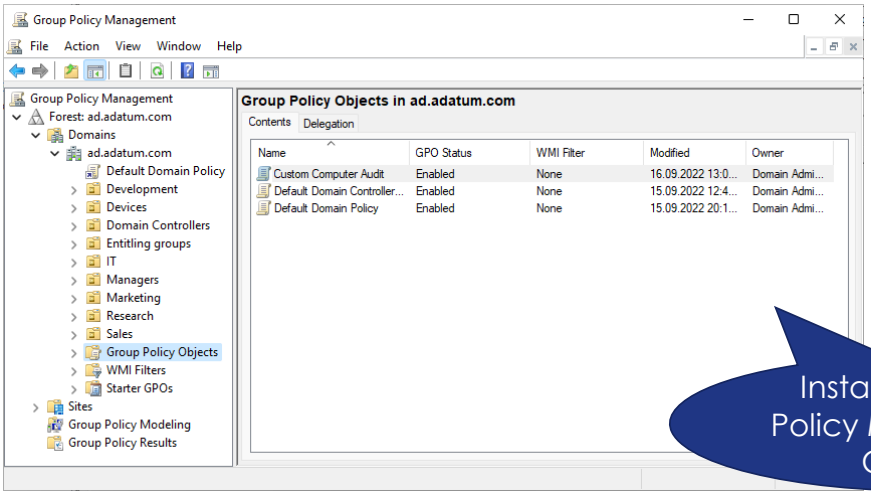
- Apply security settings
- Manage desktop application settings
- Manage Windows features
- Configure network settings

Group Policy Overview | Microsoft Learn

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Group Policy Management



Install the Group Policy Management Console

Group Policy Management Console | Microsoft Learn

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What are settings and Group Policies Objects?

A Group Policies setting defines a specific configuration change to apply to a user or a computer

A Group Policies Object is a collection of Group Policies settings that can be applied to a user, computer, or both, to enact changes

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Naming schema for GPOs

Consider a prefix for your custom GPOs, e.g., `Custom`

- Some applications create GPOs with special prefixes
- By prefixing your custom GPOs you can easily distinguish your custom GPOs from default GPOs

After the prefix, write `User` or `Computer` to distinguish between GPOs containing user settings from GPOs containing computer settings

Consider a hierarchical naming schema, e.g.,

```
Custom Computer Security Windows Firewall Remote
Desktop
```

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Creating Group Policies Objects

Desktop Experience

In the context-menu of **Group Policy Objects**, click **New**

Alternatively, you can create a GPO and link it to a container in one step:

In the context-menu of a container, click **Create a GPO in this domain, and Link it here…**

PowerShell

```
New-GPO –Name $name
```

[Working with Group Policy Objects using GPMC | Microsoft Learn]

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GPO storage



- Contains Group Policies settings
- Stores content in two locations



Group Policies container

- Stored in AD DS
- Provides version information



Group Policies template

- Stored in shared SYSVOL folder
- Provides Group Policies settings

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GPO links

To deliver settings to an object, a GPO must be linked to a container

GPOs can be linked to:

- Sites
- Domains
- OUs

GPOs cannot be linked to:

- Users
- Groups
- Computers
- System containers

A disabled link does not apply the settings of the GPO to the container

By default, GPO links are enabled

Deleting a link does not delete the GPO

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Managing GPO links

	Desktop experience	PowerShell
Creating a link	In the context-menu of a container, click **Link an Existing GPO…**	`New-GPLink ` `-Name $gpoName ` `-Target $target` `$target` is the distinguished name of a container
Removing a link	In the context-menu of a link, click **Delete**	`Remove-GPLink ` `-Name $gpoName ` `-Target $target`
List links on a container	1. Click the container 2. Explore the tabs **Linked Group Policy Objects** and **Group Policy Inheritance**	`Get-GPInheritance ` `-Target $target`

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Computer vs. user settings

Computer settings

Settings applying to the system instead of particular users

Settings which would typically require administrative rights to change the manually

Apply to computer objects in a container only

User settings

Settings applying to particular users

Settings which would typically not require administrative rights to change the manually

User settings apply to user objects in a container only

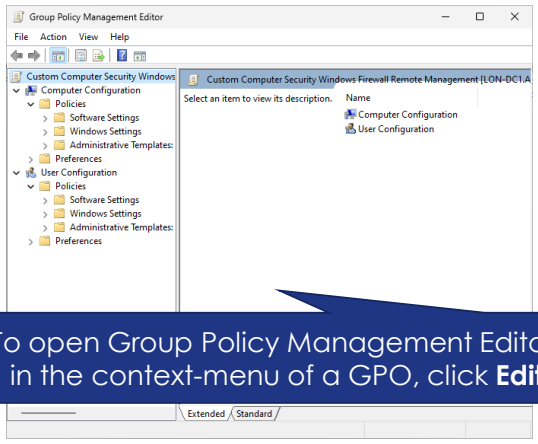
Exception: Loopback processing (not in the scope of this course)

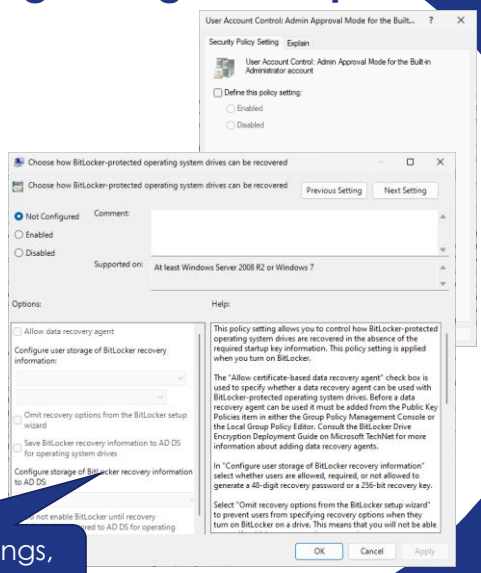
Do not mix computer and user settings in one GPO

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Configuring Group Policy settings using desktop experience





To open Group Policy Management Editor, in the context-menu of a GPO, click **Edit**

Depending on the settings, the user interface may vary

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Configuring Group Policy settings using PowerShell

Only registry-based settings (everything under Administrative Templates) can be configured using PowerShell (and registry-based Preferences, not in the scope of this course)

```
Set-GPRegistryValue `
    -Name $gpoName `
    -Key $key `
    -ValueName $valueName `
    -Type $type `
    -Value $value
```

Parameters work much like editing the registry using PowerShell

```
Get-GPRegistryValue `
    -Name $gpoName -Key $key -ValueName $valueName
Remove-GPRegistryValue `
    -Name $gpoName -Key $key -ValueName $valueName
```

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Applying GPOs

Computer settings apply at startup

User settings apply at logon

Polices refresh at regular, configurable intervals

- Default for domain controllers: every 5 minutes
- Default for domain members: every 60 – 120 minutes

Security settings refresh at least every 16 hours

Manual Policies refresh

- Local

  `gpupdate.exe`

  The `/force` parameter is required, if the user or computer was moved to a different container or group memberships have been modified.
- Remote
  - Group Policies Management Console
  - PowerShell: `Invoke-GPUpdate`

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Identifying when settings become effective

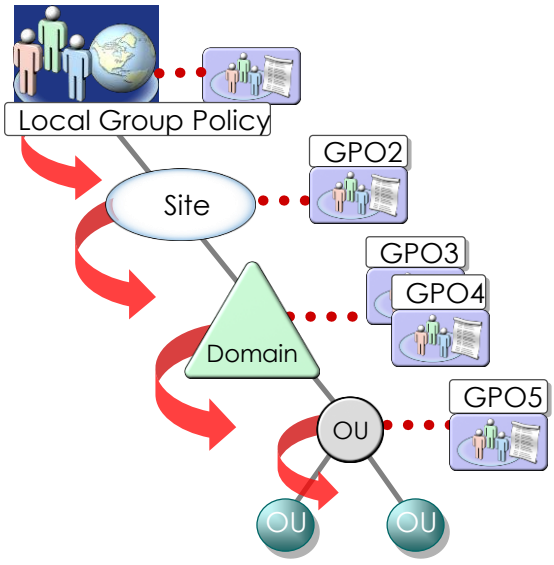
Group changes must be replicated
GPO replication must happen
Group Policies refresh must occur
Most client-side extensions (CSEs) do not reapply unchanged GPO settings
A few settings require a restart or sign out/in

- In the default configuration, these settings might require even two reboots or sing ins.
- `Gpupdate.exe` will tell you!

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Group Policies Processing Order



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Configuring GPO inheritance and precedence

The application of GPOs linked to each container results in a cumulative effect called Policies inheritance:

- Default precedence: Local → Site → Domain → OU → Child OU… (LSDOU)
- Visible on the Group Policies Inheritance tab

Link order (attribute of GPO link):

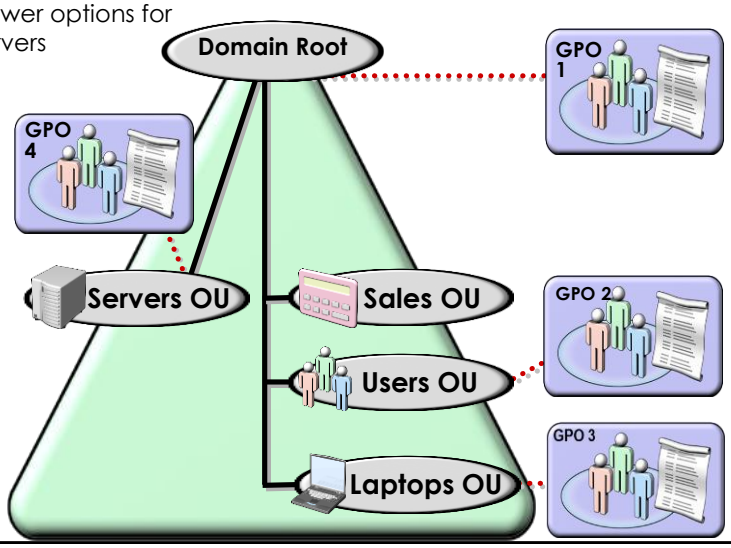
- Lower number → Higher on list → Precedence

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Discussion: Identifying Group Policies Application

GPO4 configures power options for servers



GPO1 removes access to registry tools and configures power options

GPO2 locks down desktops, removes access to Control Panel, and configures printers

GPO3 configures power options for client laptops

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What Are the Default GPOs?

There are two default GPOs:
- Default Domain Policies
- Default Domain Controllers Policies

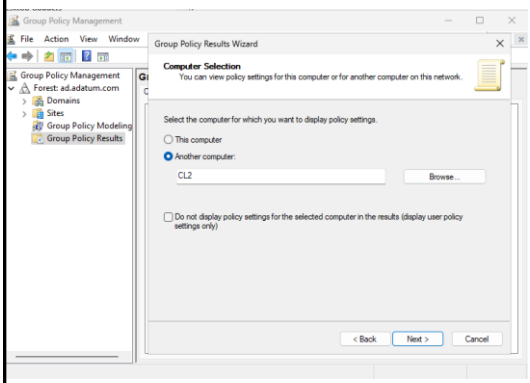
Never modify the default GPOs!
- Instead, create your own GPOs at the same level and give them a lower precedence value
- For troubleshooting purposes, you can always disable or delete your custom GPOs

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Verifying Group Policy Results

Group Policy Management Console



Locally

Create Group Policies results report for current user

```
gpresult.exe /H <path to new HTML file>
```

Create Group Policies results report for different user (administrative rights required)

```
gpresult.exe /USER <domain\user> /H
<path to new HTML file>
```

gpresult | Microsoft Learn

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Lab: Group Policies Management

1. Configure Windows Defender Firewall rules for remote administration
2. Disable the Microsoft Store
3. View Group Policy Results

Required time: 25 minutes



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More information

Course Windows Server: Gruppenrichtlinien/Group Policies » Training | ETC

Configure Firewall Port Requirements for Group Policy | Microsoft Learn

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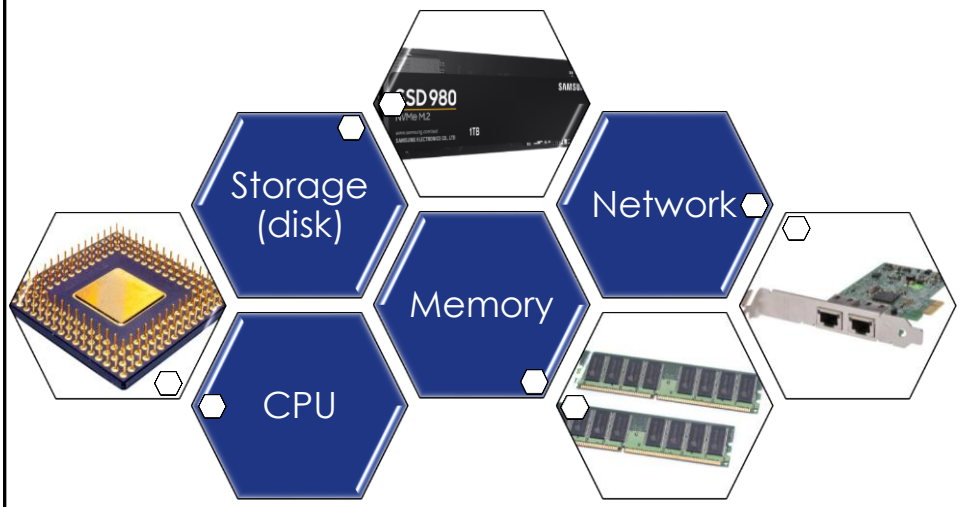
Performance monitoring

Counters, live monitoring, data collection



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Performance relevant components



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Types of servers

Type	Examples	Performance characteristics
Application server	Mail, database, virtualization (hypervisor), web, domain controller	• Heavy on CPU and memory • Heavy on small storage transactions • Caching of database pages in memory
Storage server	File server, storage pool	• Heavy on disk and network transfer • Memory used as cache
Desktop virtualization	Remote Desktop, Citrix	• Similar to application server • Memory is mainly used for applications instead of caching

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Guidelines for performance monitoring

Always measure counters of all main components

If available, measure values for unique instances instead of _Total

Avoid running performance monitoring tools on the computer under consideration, except for measuring network related counters

Consider the type of server when selecting and interpreting counters

On application servers, consider recommendations of the application supplier

Regularly create a baseline to discover trends

When troubleshooting current performance, always compare to a baseline

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Performance counters for memory

	Available Bytes	Pages/sec	Committed Bytes
Application server	Not relevant	< 50	< 1.5 * physical memory
Storage server	>= 10 % of physical memory	Not relevant	< physical memory
Desktop virtualization	>= 10 % of physical memory	< 50	< physical memory

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Performance counters for PhysicalDisk

- % Disk Time
  < 80 on average
- Current Disk Queue Length
  <= 2
- Disk Transfers/sec and Disk Bytes/sec
  Indicate the I/O performance of a disk under load
  - If these counters and Current Disk Queue Length are high, consider moving workloads to other disks
  - If these counters are low and Current Disk Queue Length is high, consider a disk upgrade

Use the same counters on LogicalDisk to identify workloads to move to other disks

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Performance counters for Processor

- % Processor Time
  < 80 % on average
  - If a single instance is > 80 % on Processor Time, but other instances not, you can try to set the low-performing application to use more threads.
  - If all instances are > 80 % on Processor Time, more CPU cores can boost performance
- % Interrupt Time
  High values may indicate a hardware related issue like a disk performance problem
- % of Maximum Frequency (in Processor information class)
  Should correlate to % Processor Time
  - Low values with high % Processor Time may indicate a thermal issue
  - High values with low % Processor Time indicate sub-optimal power management

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Performance counters for Network Adapter

- Current Bandwidth, Bytes Received/sec, Bytes Sent/sec
  - Bytes Received/sec < Current Bandwidth / 8
  - Bytes Sent/sec < Current Bandwidth / 8
- Output Queue Length
  <= 2
- Packets Received/sec
  High values may indicate a DOS attack

Packets Send Errors and Packets Receive errors

- Should be 0 or close to 0
- High values indicate a hardware-related issue
  - Defective cables
  - Wrong switch port or network interface configuration (e.g., duplex mode)
  - Defective network interface

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Performance counters for processes

Can be used to identify a misbehaving process after you identified a performance bottleneck with other counters

Typical counters in the Process class

- % Processor Time
- Page Faults/sec
- Working Set
- IO Read/Write/Other Bytes/Operations per sec

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Considerations for monitoring VMs

Virtual machines must be assigned sufficient resources for their workload

If multiple virtual machines run on a host, ensure the host has enough resources

Resources are shared, so performance of one virtual machine can affect the performance of others

You must remember to monitor the resource utilization on the host as well as the guests

Use PowerShell VMRessource Metering cmdlets (*-VMResourceMetering)

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Discussion

Assume an application server has not enough memory.
For which performance counters will you see abnormal values?
What values do you expect from these counter?
Required time: 5 minutes



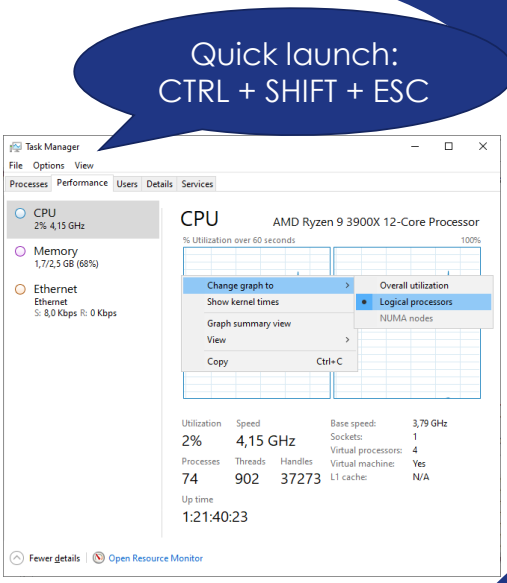
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Task Manager

Use for instant troubleshooting
Counters on tab **Performance**
- CPU Utilization: % Processor Time
  Change graph to **Logical Processors**
- Memory Committed
- Ethernet Bits sent/received per second

Counters on tab **Details**
- CPU usage per process
- Memory usage per process

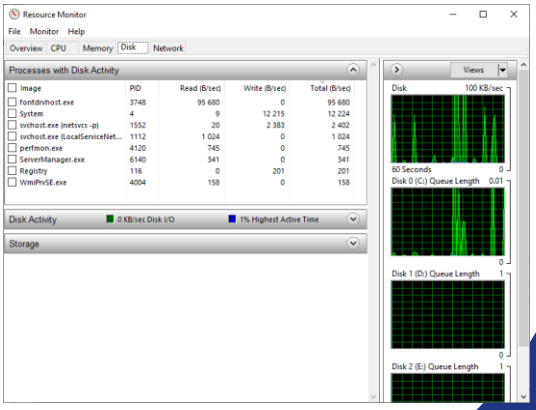


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Resource Monitor (resmon.exe)

Resource Monitor provides an in-depth look at the real-time performance of your server
Most important counters are included
Counters per process are included
Use for instant troubleshooting



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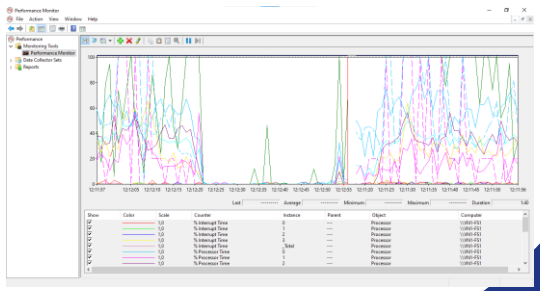
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Performance snap-in in MMC (perfmon.exe)

Performance Monitor enables you to view current performance statistics, or to view historical data gathered through the use of data collector sets

Data collector sets enable you to gather performance-related and other system statistics for analysis and can contain

- Performance counters
- Event trace data
- System configuration information

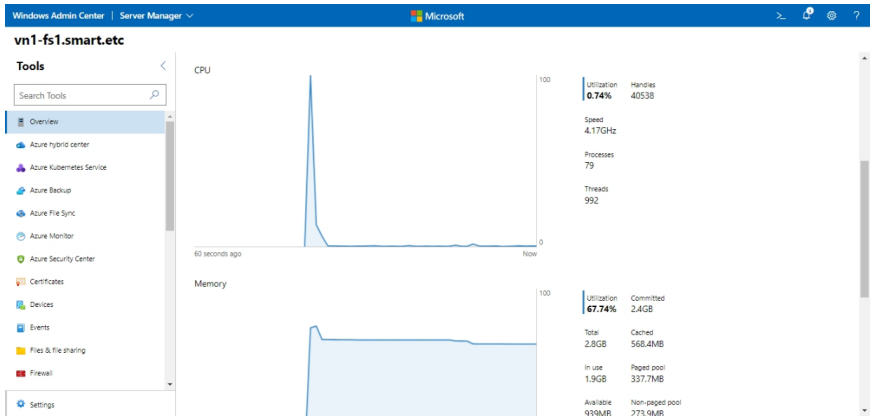


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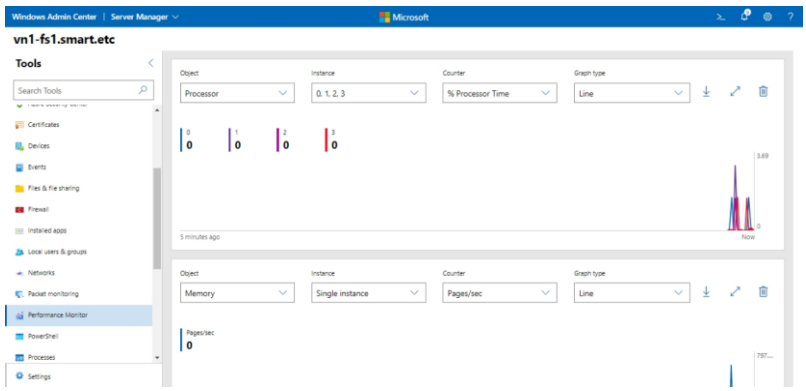
Windows Admin Center (Overview)

Live monitoring
Similar to Task Manager



Windows Admin Center (Performance Monitor)

Live monitoring
Data can be downloaded as CSV
Use workspaces to save sets of counters



Lab: Monitor performance

1. Monitor performance under low memory conditions
2. Monitor performance after a memory upgrade
3. Use a performance alert

Required time: 45 minutes



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Licensing

Windows Server editions, licensing, virtualization rights, Azure hybrid use benefit



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Windows Server editions

Windows Server Datacenter
Windows Server Standard
Windows Storage Server
Microsoft Hyper-V Server

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Selecting a suitable Windows Server edition

Windows Server 2022 Edition	Ideal for	Licensing model	CAL requirements[1]
Datacenter[2]	Highly virtualized datacenters and cloud environments	Core-based	Windows Server CAL
Standard[2]	Physical or minimally virtualized environments	Core-based	Windows Server CAL
Essentials	Small businesses with up to 25 users and 50 devices	Specialty servers (server license)[3]	No CAL required

[1] CALs are required for every user or device accessing a server. See the Product Use Rights for details.

[2] Datacenter and Standard edition pricing is for 16 core licenses.

[3] Up to 10 cores and 1 VM on single-socket servers. Windows Server Essentials is available through our OEM Server Hardware partners.

[4] Pricing is shown in USD and may vary from country to country. Please contact your Microsoft representative for a quote.

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Feature Comparison – Standard/Datacenter

Feature support key
● Feature available ○ Feature not available ◑ Limited feature

Feature	Standard edition	Datacenter edition
Core Windows Server functionality	●	●
Hybrid integration	●	●
OSEs*/Hyper-V isolated containers	2 [1]	Unlimited
Windows Server containers	Unlimited	Unlimited
Storage Replica [2]	◑	●
Software-defined networking	○	●
Software-defined storage	○	●

Windows Server 2022 Licensing & Pricing | Microsoft

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Licensing of Windows Server

Licensing of Windows Server is based on physical cores

Minimum license requirements

- All physical cores must be licensed
- 8 core licenses per processor
- 16 core licenses per server

In addition to a server license, you will need a Windows Server Client Access License (WS CAL)

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Licensing of Windows Server 2019

Physical Cores per Processor	2	4	8	10	12	14	16	Processors per server
Number of 2-core packs needed	8	8	8	8	8	8	8	1 processor
	8	8	8	10	12	14	16	2 processors
	16	16	16	20	20	24	32	3 processors

Windows Server | Licensing Guide

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Virtualization Rights

Standard Edition provides rights for up to two OSEs or Hyper-V containers

- When all physical cores in the server are licensed. For every two additional VMs, all the cores in the server have to be licensed again

Datacenter Edition provides unlimited rights for OSEs or Hyper-V containers

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Azure Hybrid Use Benefit

Use on-premise licenses for Azure VMs – the compute rate is equal to the Linux rate

Run 2 instances with up to 8 cores, or one instance with up to 16 cores

With Server Datacenter you can use your license on-premise and in the cloud at the same time

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Practice: Display licensing information

Display help for the license manager tool and display the licensing information of VN1-SRV5.

Required time: 5 minutes



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